

Polystyrene nanoparticles trigger aberrant condensation of TDP-43 and amyotrophic lateral sclerosis-like symptoms

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Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by the dysfunction and progressive death of cerebral and spinal motor neurons. Preliminary epidemiological research has hinted at a relationship between environmental risks and the escalation of ALS, but the underlying reasons remain mostly mysterious. Here we show that nanosize polystyrene plastics (PS) induce ALS-like symptoms and illustrate the related molecular mechanism. When exposed to PS, cells endure internal oxidative stress, which leads to the aggregation of TAR DNA-binding protein 43 kDa (TDP-43), triggering ALS-like characteristics. In addition, the oxidized heat shock protein 70 fails to escort TDP-43 back to the nucleus. The cytoplasmic accumulation of TDP-43 facilitates the formation of a complex between PS and TDP-43, enhancing the condensation and solidification of TDP-43. These findings are corroborated through *in silico* and *in vivo* assays. Altogether, our work illustrates a unique toxicological mechanism induced by nanoparticles and provides insights into the connection between environmental pollution and neurodegenerative disorders.

Plastic products are broadly used nowadays, and the ensuing plastic pollution has become a global problem. It is estimated that approximately 11 billion tons of plastic waste will have been placed in the environment by 2025¹. Plastics may undergo continuous fragmentation and shred into micrometre- or even nanometre-scale particles (termed microplastics and nanoplastics (MNPs)) often detected in water, soil and air². Specifically, airborne MNPs can float and transport in the atmosphere; previous studies have indicated that plastic fibres can be found in human lung tissue, suggesting that airborne MNPs are inhalable³.

Similar to other small particles, MNPs, upon their uptake, can reach the brain and exert a wide range of neurotoxic effects^{4,5}; however,

there is still an urgent demand to further illuminate the neurotoxic mechanism(s) of MNPs. Mounting evidence indicates that neurodegenerative diseases (NDDs) are highly associated with ambient air pollutants and ultrafine particle exposure⁶. Recent epidemiological studies have progressively linked airborne particle exposure to the incidence of amyotrophic lateral sclerosis (ALS)⁷.

ALS is a lethal NDD that attacks neurons, primarily affecting the human motor system⁸. Approximately 10% of patients with ALS are considered to have familial traits or genes, and 90% of ALS cases are sporadic. Meanwhile, environmental factors may trigger motor neuron degeneration over time⁹. TAR DNA-binding protein 43 (TDP-43), a predominant nuclear RNA- and DNA-binding protein, is a typical

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pathological hallmark in ALS cases¹⁰. In a normal state, TDP-43 is mainly restrained in the nucleus and exerts multiple cellular functions in RNA regulation¹¹; whereas under pathological or stressed conditions, TDP-43 suffers post-translational modifications, which causes its cytoplasmic accumulation and aggregation¹². Indeed, TDP-43 mislocalization and aggregation are prevalent in various neurological diseases, including ALS (~97% of patients), frontotemporal dementia (~50% of patients) and Alzheimer's disease (30–50% of patients)¹³. In patients with ALS, pathological TDP-43 spreads from projection neurons of the agranular motor cortex and somatomotor neurons of the brainstem/spinal cord to other regions¹⁴. In recent years, evidence has revealed that in response to cellular stress, physiological TDP-43 is driven by liquid–liquid phase separation (LLPS), a process of biomolecules moving from a dilute aqueous phase to a condensed demixed phase, to form biomolecular condensates and perform its physiological functions¹⁵.

During cellular stress, a portion of TDP-43 participates in the assembly of stress granules (SGs), which are cytoplasmic membrane-less organelles, and protects TDP-43 from disease-related abnormal phosphorylation¹⁶. Heat shock protein 70 (Hsp70) can maintain the normal physiological LLPS behaviour of nuclear or cytoplasmic TDP-43 by binding to TDP-43 (ref. 17). As a highly sensitive protein to external stimuli, TDP-43 may respond to the stress caused by MNPs entering cells as exogenous substances, thus forming intracellular condensates under the drive of LLPS¹⁸. Owing to their high surface free energy, nanoparticles tend to adsorb encountered biomolecules¹⁹; therefore, it is worthwhile to investigate the spatial location relationship between internalized MNPs and TDP-43 condensates and how MNP adsorption affects the localization and assembly of cytoplasmic TDP-43 condensates.

Polystyrene plastics (PS), thermoplastic polymers with high transparency and durability, constitutes an important component of MNP pollution²⁰. Here we report the assembly of TDP-43 LLPS and the aberrant phase transitions of TDP-43 condensates upon exposure to nanosize PS. Furthermore, PS adsorbs cytoplasmic TDP-43 to form a complex that promotes TDP-43 condensation and solidification; the consistent result was confirmed by molecular docking simulation. We also conducted an in vivo animal study by intranasal exposure of PS to mice and found that PS-treated mice showed ALS-like abnormal behaviours and liquid-droplet-like TDP-43 formation in the brain region. These findings have significant implications for risk assessment of PS exposure and the prevention of health outcomes.

Dynamic assembly and phase transition of TDP-43 condensates

The characterization of all PS is shown in Supplementary Fig. 1. First, we exposed plain PS to HeLa cells, and found that transgenic or endogenous nuclear TDP-43 in these cells changed from a normal diffuse pattern to a distinct granular appearance (Fig. 1a and Extended Data Fig. 1a).

Fig. 1 | PS regulates the dynamic assembly and liquid-to-solid phase transition of nuclear TDP-43 condensates. **a**, Representative images and the fluorescence intensity profile along the indicated blue line of HeLa cells forming ^{EGFP}TDP-43 granules (arrows) at 6 h after the indicated concentrations of PS treatment. **b**, Confocal microscopy delayed capture image showing the fusion of two ^{EGFP}TDP-43 droplets in PS-exposed HeLa cells. The yellow and red arrowheads point to two separated droplets before fusion, and the blue arrowheads point to the fused droplet. PS, 100 µg ml⁻¹, 12 h. **c**, FRAP representative images of nuclear ^{EGFP}TDP-43 condensates in PS-exposed cells. PS, 100 µg ml⁻¹, 6 h. **d**, Nuclear ^{EGFP}TDP-43 condensates formation. PS (0, 25 µg ml⁻¹, 50 µg ml⁻¹, 100 µg ml⁻¹) was exposed to HeLa cells for 6 h, 12 h and 24 h. The red box highlights a highly condensed state. **e**, Nuclear ^{EGFP}TDP-43 condensates number per cell in **d** (*n* = 15 cells). Data are displayed as median ± interquartile range. Statistical analysis was performed through one-sided two-way ANOVA with Tukey's multiple comparison tests (*P* < 0.0001; NS, *P* > 0.05; Ctrl, 6 h versus PS 25 µg ml⁻¹, 6 h, *P* = 0.0089; PS 25 µg ml⁻¹, 6 h versus PS 50 µg ml⁻¹, 6 h, *P* = 0.0108; PS 25 µg ml⁻¹, 12 h versus

A proper intracellular environment (for example, crowding) and a certain concentration (>30 µM) of TDP-43 protein is essential for PS to trigger intracellular TDP-43 granular formation (Supplementary Fig. 2). The granular in the nucleus has a spherical shape and a certain degree of bulge on the surface of the nucleus (Supplementary Fig. 3 and Supplementary Videos 1 and 2). The biomolecular condensates usually show a liquid-like feature, continuously and dynamically exchanging materials with their surroundings²¹. Using enhanced green fluorescent protein (EGFP) as the tag for TDP-43 (^{EGFP}TDP-43), we analysed the fusion event and fluorescence recovery after photobleaching (FRAP) of TDP-43 to confirm the dynamic of PS-induced TDP-43 granules. Red and yellow arrowheads represent two ^{EGFP}TDP-43 droplets before the fusion, and blue arrowheads represent the completion of fusion (Fig. 1b and Supplementary Video 3). The area of the granules after the fusion is significantly larger than that before the fusion (Supplementary Fig. 4a), which implies the liquid-like property of spherical TDP-43 granules. FRAP of ^{EGFP}TDP-43 granules revealed rapid recovery, reaching ~50% of initial intensity in approximately 75 s (Fig. 1c, Supplementary Fig. 4b and Supplementary Video 4).

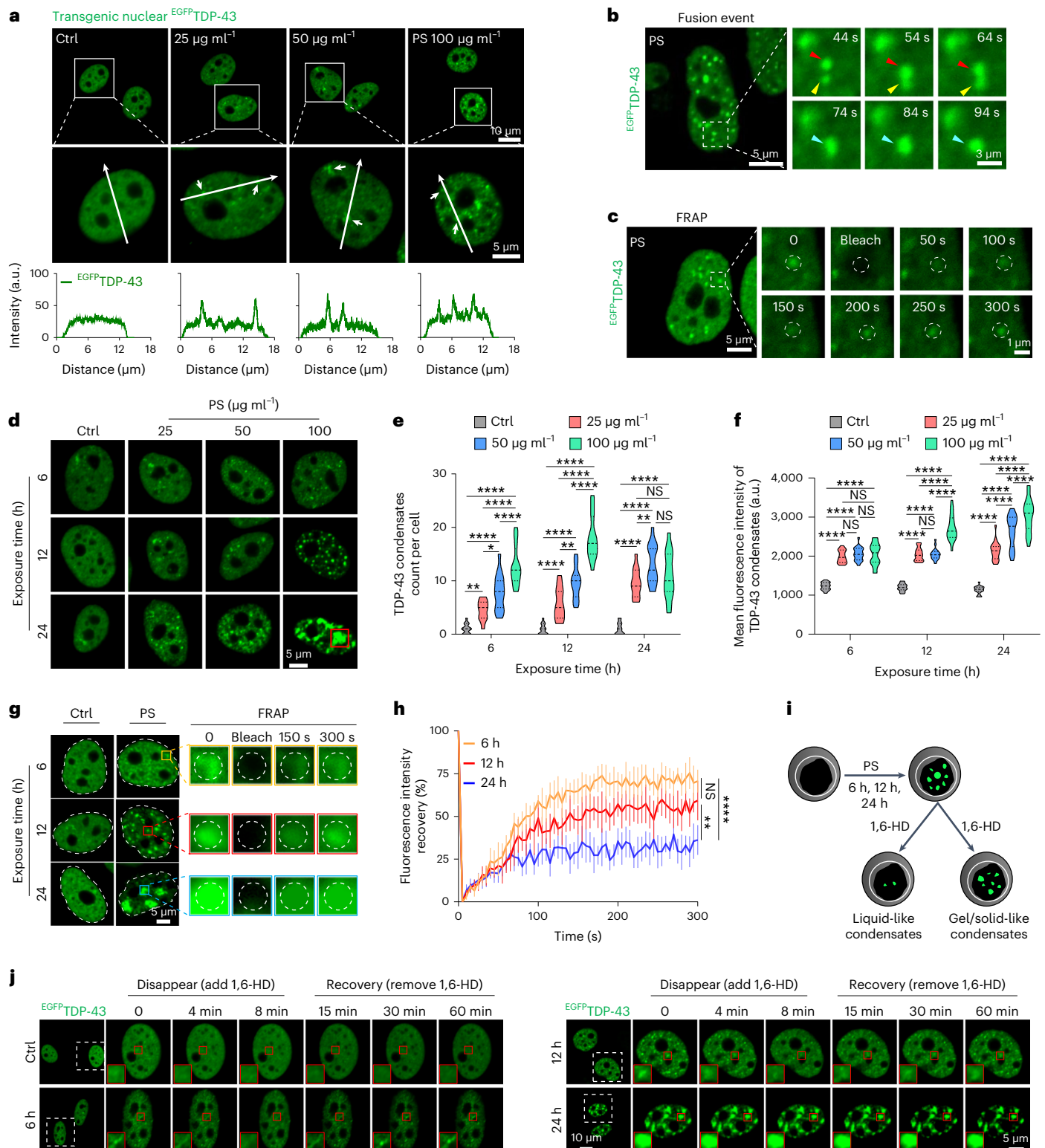
Oxidative stress has been reported to induce TDP-43 aggregation in mouse and human-origin neuronal cells²². Thus, we next examined whether oxidative stress contributes to PS-induced nuclear TDP-43 condensate assembly. As expected, PS (100 µg ml⁻¹) exposed to cells for 6 h induced nuclear TDP-43 condensates formation, but the pre-treatment with *N*-acetyl-L-cysteine (NAC) remarkably inhibited the assembly of TDP-43 condensates in the nucleus (Extended Data Fig. 2a,b). The counts of nuclear TDP-43 condensates per cell were remarkably reduced after NAC pre-treatment (Extended Data Fig. 2c). Therefore, cellular oxidative stress caused by PS exposure may be essential in triggering nuclear TDP-43 LLPS. Although NAC completely inhibited the reactive oxygen species (ROS) level, it failed to block all nuclear TDP-43 condensates formation (Extended Data Fig. 2b, control (Ctrl) versus PS + NAC, no significance; Extended Data Fig. 2c, Ctrl versus PS + NAC, *P* < 0.01). This result implied that other mechanisms may apply. Eukaryotic initiation factor-2α (eIF2α) is an essential intracellular signalling protein that responds to cellular stress. The phosphorylation of eIF2α promotes the formation of SGs and aberrant aggregation of NDD-related proteins²³. As shown in Extended Data Fig. 2d,e, eIF2α responded to intracellular stress caused by PS exposure, leading to elevated p-eIF2α expression levels. Mitochondrial dysfunction potentially acts as a point of convergence for TDP-43 proteinopathy²². JC-1 staining assay indicated the compromised mitochondria function (Extended Data Fig. 2f,g and Supplementary Fig. 5). Meanwhile, propidium iodide, acridine orange and magic red staining assays indicated that PS caused cell membrane damage and increased lysosomal membrane permeability (Extended Data Fig. 2f,g and Supplementary Fig. 5).

The dynamic assembly of nuclear TDP-43 condensates driven via LLPS was dominated by the degree of stress²⁴. Specifically, aberrant LLPS causes the biomolecular condensates to undergo a liquid-to-solid

PS 50 µg ml⁻¹, 12 h, *P* = 0.0032; PS 25 µg ml⁻¹, 24 h versus PS 50 µg ml⁻¹, 24 h, *P* = 0.0073). **f**, Mean fluorescence intensity of nuclear ^{EGFP}TDP-43 condensates (*n* = 20 cells, 3 condensates per cell). Data are displayed as median ± interquartile range. Statistical analysis was performed through one-sided two-way ANOVA with Tukey's multiple comparison tests (*P* < 0.0001; NS, *P* > 0.05; PS 25 µg ml⁻¹, 6 h versus PS 50 µg ml⁻¹, 6 h, *P* = 0.9995; PS 25 µg ml⁻¹, 6 h versus PS 100 µg ml⁻¹, 6 h, *P* > 0.9999; PS 50 µg ml⁻¹, 6 h versus PS 100 µg ml⁻¹, 6 h, *P* > 0.9999; PS 25 µg ml⁻¹, 12 h versus PS 50 µg ml⁻¹, 12 h, *P* > 0.9999). **g**, FRAP analysis of nuclear ^{EGFP}TDP-43 condensates induced by PS (100 µg ml⁻¹). **h**, The recovery of fluorescence intensity in photobleached ^{EGFP}TDP-43 condensates in **g** (*n* = 6 bleached condensates; graph represents mean ± s.d.). Statistical analysis was performed through one-sided two-way ANOVA with Tukey's multiple comparison tests (PS 6 h versus PS 12 h, *P* = 0.1567; PS 6 h versus PS 24 h, *P* < 0.0001; PS 12 h versus PS 24 h, *P* = 0.0085). **i**, Schematic diagram of 1,6-HD-dissolved liquid-like TDP-43 condensates. **j**, Representative images of ^{EGFP}TDP-43 condensates with 1,6-HD (3%) treatment and recovery. **P* < 0.05, ***P* < 0.01, *****P* < 0.0001 (e,f,h).

phase transition and highly condense into irreversible pathological fibrils associated with neurodegenerative disorders such as ALS²⁵. The dynamic assembly of nuclear TDP-43 condensates stimulated by PS was investigated, and the surrounding TDP-43 protein molecules continuously accumulated to form irregular and large-size TDP-43 condensates (Fig. 1d). Interestingly, the number of nuclear TDP-43 condensates per cell is positively correlated with the time and concentration of PS exposure at the early stages (6 h and 12 h). However,

a decreased number of TDP-43 condensates in the nucleus occurred over time (Fig. 1e; 100 $\mu\text{g ml}^{-1}$ PS, 12 h versus 24 h, $P < 0.0001$), suggesting a highly condensed state of TDP-43. Meanwhile, the mean fluorescence intensity of the nuclear TDP-43 condensates continuously increased with prolonged PS exposure (Fig. 1f). Using red fluorescent PS (R^{F} PS) as a surrogate, we found a time-dependent increase of R^{F} PS fluorescence in HeLa cells during 24 h of exposure (Supplementary Fig. 6), indicating that the ascending stress facilitates the dynamic assembly



of nuclear TDP-43 condensates. Highly spontaneous self-assembly of TDP-43 via aberrant LLPS yields solidified condensates²⁴. To further verify the liquid-to-solid transition of TDP-43 condensates induced by PS, we performed FRAP analysis and found that large-size TDP-43 condensates (24 h group) had the slowest recovery rate (Fig. 1g,h). As decreased dynamics of nuclear TDP-43 condensates was observed, we asked whether TDP-43 condensates transformed into an aberrant phase transition end product. 1,6-Hexanediol (1,6-HD; a specific inhibitor of biomolecular condensates) treatment rapidly dissolved PS-induced (6 h and 12 h exposure) TDP-43 condensates in HeLa cells, while the morphology of condensates was recovered after 1,6-HD removal (Fig. 1i,j). This reversible phenomenon indicated that the early nuclear TDP-43 condensates were still liquid-like. However, the late-stage condensates (24 h) were resistant to 1,6-HD, indicating a solid-like state of condensates.

Solid-like nuclear TDP-43 condensates formation

Abnormal phosphorylation of the low-complexity domain in accumulated TDP-43 leads to increased fibrillization²⁶. TDP-43 phosphorylation at the S409/410 site is an inducement for its aggregation²⁷. Although TDP-43 condensates were found in all PS-treated groups, the red nuclear foci of phosphorylated TDP-43 s409/410 (pTDP-43 s409/410) were observed in only the 12 h and 24 h groups, not in 6 h group (Extended Data Fig. 3a,b). PS exposure promoted the expression of pTDP-43 in the nucleus in a time-dependent fashion (Extended Data Fig. 3c,d). Cytoplasmic inclusion bodies, aggresomes or amyloid fibrils containing pTDP-43 were regarded as significant markers of ALS²⁸. Indeed, PS-induced solid-like and irregular TDP-43 condensates were observed accumulating in the intranuclear region and forming amyloid-like clusters, as confirmed by an amyloid-specific dye AmyTracker 680 (AmyT) (Extended Data Fig. 3e). The number of AmyT-positive cells increased over time (Extended Data Fig. 3f), implying the depletion of storage capacity of the nuclei for misfolded proteins, resulting in the transition of liquid TDP-43 into a solid, aggregated state, and ultimately fibrils²⁹.

Amyloid formation is one of the major causes of cytotoxicity in degenerative diseases³⁰. Therefore, we sought to retard PS-induced solid-like TDP-43 formation by inhibiting TDP-43 assembly, which might impede amyloid generation and cytotoxicity. We introduced a helix breaker, TDP-43 A326P mutation, into the conserved region (316–346 aa)³¹, and found that A326P mutation almost completely inhibited TDP-43 LLPS (Extended Data Fig. 3g). The disrupted helical structure of TDP-43 does not affect the cell viability of transfected HeLa cells within 24 h (Supplementary Fig. 7). However, upon PS treatment, the loss of cell viability was further exacerbated in the A326P variant than in the wild type in the 6 h and 12 h groups. In comparison, A326P mutation partially restored the cell viability in the 24 h group (Extended Data Fig. 3h). This result implies that intranuclear TDP-43 condensates formed during the early and middle stages of PS exposure may play a role in resisting exogenous stress. In parallel, a previous report also demonstrated that stress-induced phase-separated liquid-like TDP-43 nuclear bodies can attenuate cytotoxicity³⁸. Nevertheless, continuous PS exposure leads to amyloid TDP-43, which is no longer protective and may instead be cytotoxic. Conversely, A326P mutation inhibits the formation of not only TDP-43 condensates but also amyloid TDP-43.

Solid-like nuclear TDP-43 condensates formation by oxidizing Hsp70

Hsp70 is a molecular chaperone of nuclear and cytoplasmic TDP-43 condensates³², which responds to cellular stress and assists in recovering proteostasis²⁹. Cytoplasmic Hsp70 with red fluorescent protein tag (^{RF}Hsp70) shuttled to the nucleus and recruited into TDP-43 condensates upon PS challenge (Fig. 2a). Interestingly, Hsp70 expression showed an inverted U-shape pattern in both immunofluorescence

and western blotting assays. Specifically, PS-induced cellular stress elevated Hsp70 expression at 6 h and 12 h, while abnormally decreasing Hsp70 expression was found at 24 h (Fig. 2a–c). Oxidative insult can cause structural modifications of Hsp70, such as the formation of inter- or intramolecular disulfide bonds³³. As expected, an ~140 kDa band (dimer) was identified in PS-treated groups in a time-dependent manner. β -Mercaptoethanol (β Me), a potent reducing agent for cleaving protein disulfide bonds, completely inhibited Hsp70 dimer formation (Fig. 2d). These data indicated that Hsp70 rapidly responded to PS at an early stage; however, sustained PS treatment oxidized Hsp70 and potentially impaired its chaperone role.

To further understand the ‘dynamic keeper’ role of Hsp70, we determined the impact of Hsp70 expression on PS-triggered TDP-43 condensates. To achieve this, *HSPA1A* (encoding Hsp70) was down-regulated using small interference RNA (siRNA) (Supplementary Fig. 8). si-*HSPA1A* reduced the assembly of PS-triggered nuclear TDP-43 condensates (Fig. 2e,f), confirming a specific function of Hsp70 in assisting the assembly of nuclear TDP-43 condensates²⁴. *HSPA1A* knockdown cells are more susceptible to PS challenge, resulting in an elevated cell viability loss with prolonged PS exposure (Fig. 2g). FRAP assay also confirmed that *HSPA1A* knockdown affects the dynamics of PS-induced nuclear TDP-43 condensates in the 12 h and 24 h groups (Fig. 2h and Supplementary Fig. 9).

TDP-43 cytoplasmic location promotes its binding with PS

TDP-43 is a mainly located in the nucleus; however, it can shuttle between the nucleus and cytoplasm. The mislocalization of TDP-43 could drive early pathology of ALS³⁴; thus we speculate that the loss of Hsp70 function yields the accretion of cytoplasmic TDP-43 in PS-exposed cells. We also question whether there is a physical interaction between internalized PS and cytoplasmic TDP-43.

Using arsenite as a positive control, we found that nuclear ^{EGFP}TDP-43 was translocated to the cytoplasm and subsequently recruited to SGs in both PS and arsenite-treated cells (Fig. 3a–c, Extended Data Fig. 4a and Supplementary Videos 5–7). SGs were defined using an SG marker T-cell-restricted intracellular antigen 1 (TIA-1)-related protein (TIAR)³⁵. Three-dimensional rendering revealed that TDP-43 was sporadically scattered in the cytoplasm (Fig. 3a, yellow arrow). Furthermore, PS-exposure-triggered cytoplasmic mislocalization of TDP-43 was confirmed (Extended Data Fig. 4b–d). Thus, we hypothesized that PS’s direct contact with cytoplasmic TDP-43 influences the recruitment of TDP-43 to SGs. To verify this, we treated HeLa cells with ^{RF}PS and then observed the spatial location of ^{RF}PS and cytoplasmic ^{EGFP}TDP-43 by confocal microscope. As expected, the orange foci clearly showed co-localization of ^{RF}PS and ^{EGFP}TDP-43 in the cytoplasm (Fig. 3d,e). In addition, the spatial location relationship of the PS@TDP-43 complexes was further investigated using three-dimensional rendering (Fig. 3f and Supplementary Video 8). Fluorescence intensity distribution analysis indicated that the PS@TDP-43 complexes have two modes, an encapsulation mode (Fig. 3g, top) and a proximity mode (Fig. 3g, bottom). These two spatial location patterns implied direct contact between ^{RF}PS and ^{EGFP}TDP-43 condensates in the cytoplasm. Notably, although SG assembly facilitates the transition of TDP-43 from soluble to liquid-like forms, it was previously identified that TDP-43 can aggregate independent of SG formation under mild chronic stress conditions³⁶.

To prove that PS adsorbs adjacent TDP-43 in the cytoplasm, we first incubated PS with recombinant TDP-43–maltose-binding protein-histidine (TDP-43–MBP-His) protein and semi-quantified the corresponding adsorption with a western blotting assay (Supplementary Fig. 10a). The amount of adsorbed TDP-43–MBP-His protein increased proportionately to PS. We further incubated PS with HeLa cell lysates and confirmed the existence of TDP-43 on the

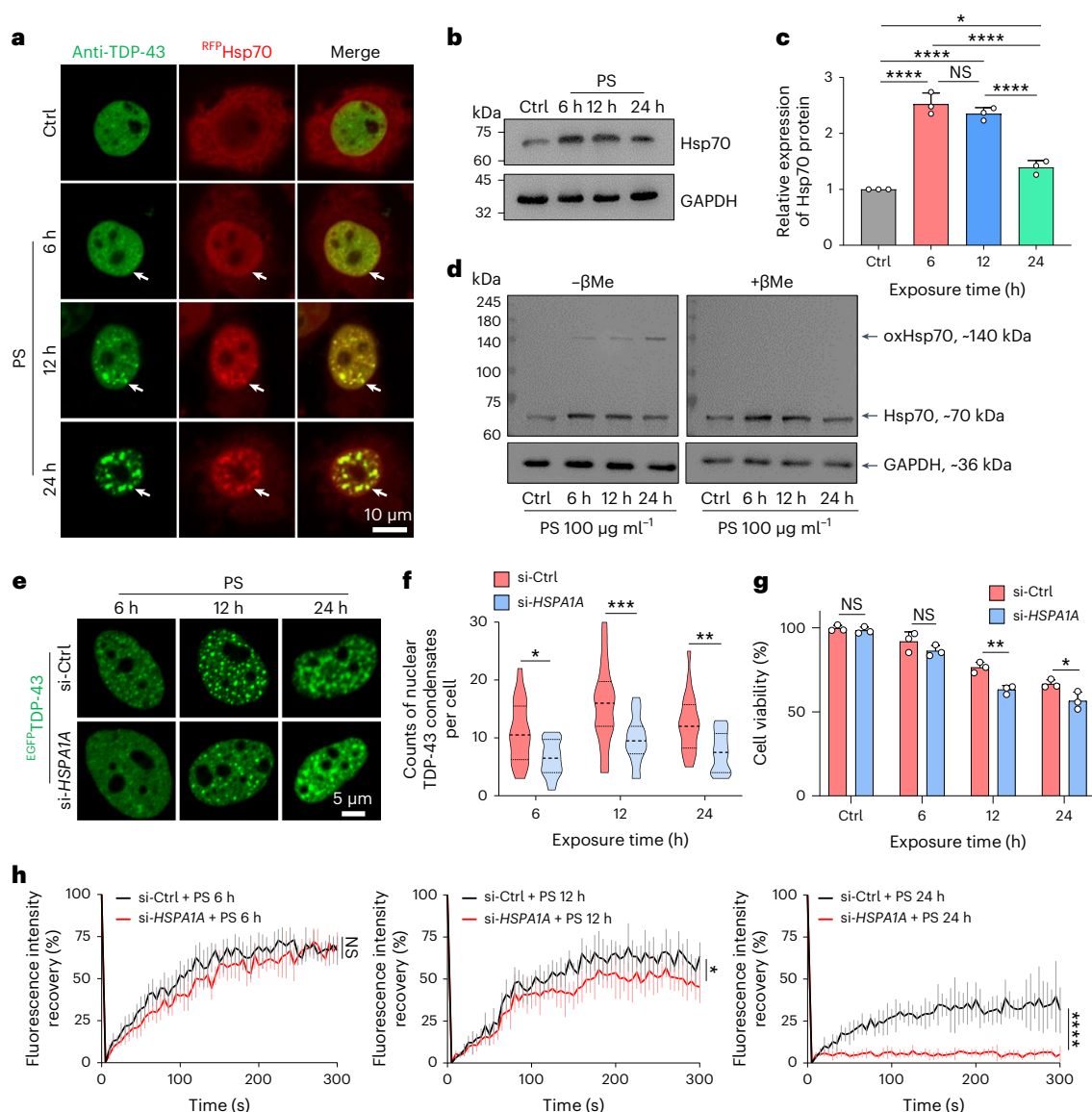


Fig. 2 | PS accelerates solid-like nuclear TDP-43 condensates formation by oxidizing chaperone Hsp70. HeLa cells were exposed to PS (100 µg ml⁻¹). **a**, Arrows indicate the co-localization of RFP-Hsp70 with TDP-43 condensates. **b**, Representative images of Hsp70 level in HeLa cells after PS treatment. Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) was used as an internal control. **c**, Quantification of Hsp70 expression in **b**. Data are expressed as mean ± s.d. ($n = 3$), as determined by one-way ANOVA with Tukey's multiple comparison tests (Ctrl versus PS 6 h, 12 h and PS 6 h versus PS 24 h and PS 12 h versus PS 24 h, $P < 0.0001$; Ctrl versus PS 24 h, $P = 0.0241$; PS 6 h versus PS 12 h, $P = 0.4093$). **d**, Representative images of redox western blotting under non-reducing (-βMe) or reducing (+βMe) conditions. oxHsp70, oxidized Hsp70 dimer. **e**, The effect of si-HSPA1A on the assembly of nuclear TDP-43 condensates. **f**, The average count of nuclear TDP-43 condensates per cell in **e**, $n = 20$ cells per group. Data are displayed as median ± interquartile range, as determined by two-

way ANOVA with Bonferroni's multiple comparison tests (PS 6 h, si-ctrl versus si-HSPA1A, $P = 0.0103$; PS 12 h, si-ctrl versus si-HSPA1A, $P = 0.0004$; PS 24 h, si-ctrl versus si-HSPA1A, $P = 0.0050$). **g**, CCK-8 assay showing that si-HSPA1A promoted cytotoxicity in PS-exposed cells at 12 h and 24 h. Data are expressed as mean ± s.d. ($n = 3$), as determined by two-way ANOVA with Bonferroni's multiple comparison tests (PS 12 h, si-ctrl versus si-HSPA1A, $P = 0.0014$; PS 24 h, si-ctrl versus si-HSPA1A, $P = 0.0130$; NS, $P > 0.05$). **h**, FRAP analysis was performed to evaluate the effect of si-HSPA1A on the dynamics of nuclear EGFP-TDP-43 condensates ($n = 6$ bleached condensates; graph represents mean ± s.d.). Statistical analysis was performed through one-sided two-way ANOVA with Tukey's multiple comparison tests (si-ctrl + PS 6 h versus si-HSPA1A + PS 6 h, $P > 0.9999$; si-ctrl + PS 12 h versus si-HSPA1A + PS 12 h, $P = 0.0116$; si-ctrl + PS 24 h versus si-HSPA1A + PS 24 h, $P < 0.0001$). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$ (**c,f,g**).

surface of PS (Supplementary Fig. 10b). Therefore, the construction of PS@TDP-43 complex may hinder the recruitment of TDP-43 to SGs.

SGs may act as 'safe compartment' to protect cytoplasmic TDP-43 from disease-related phosphorylation; therefore, cytoplasmic TDP-43 that separated from SGs was considered a potential client for pathological aggregation and oligomerization³⁷. Given the nature of PS adsorption, we hypothesized that the development of the PS@TDP-43 complex destroyed the dynamics of cytoplasmic TDP-43.

After RFP-PS treatment (100 µg ml⁻¹, 12 h), we found both 'RFP-PS-attached' complex-state EGFP-TDP-43 (RFP-PS@EGFP-TDP-43; Fig. 4a,b, blue box) and 'RFP-PS-free' independent cytoplasmic EGFP-TDP-43 (Fig. 4a,b, yellow box). Surprisingly, the dynamics of RFP-PS@EGFP-TDP-43 were markedly lower than that of EGFP-TDP-43 (Fig. 4c).

To further study the dynamics of RFP-PS@EGFP-TDP-43 in the cytoplasm, we used EGFP-TDP-43 NLS^{mut} plasmid to construct EGFP-TDP-43 protein with site-directed mutagenesis, which lost its nuclear localization function³⁸ (Fig. 4d). When purple fluorescent PS (P^{PS}) was

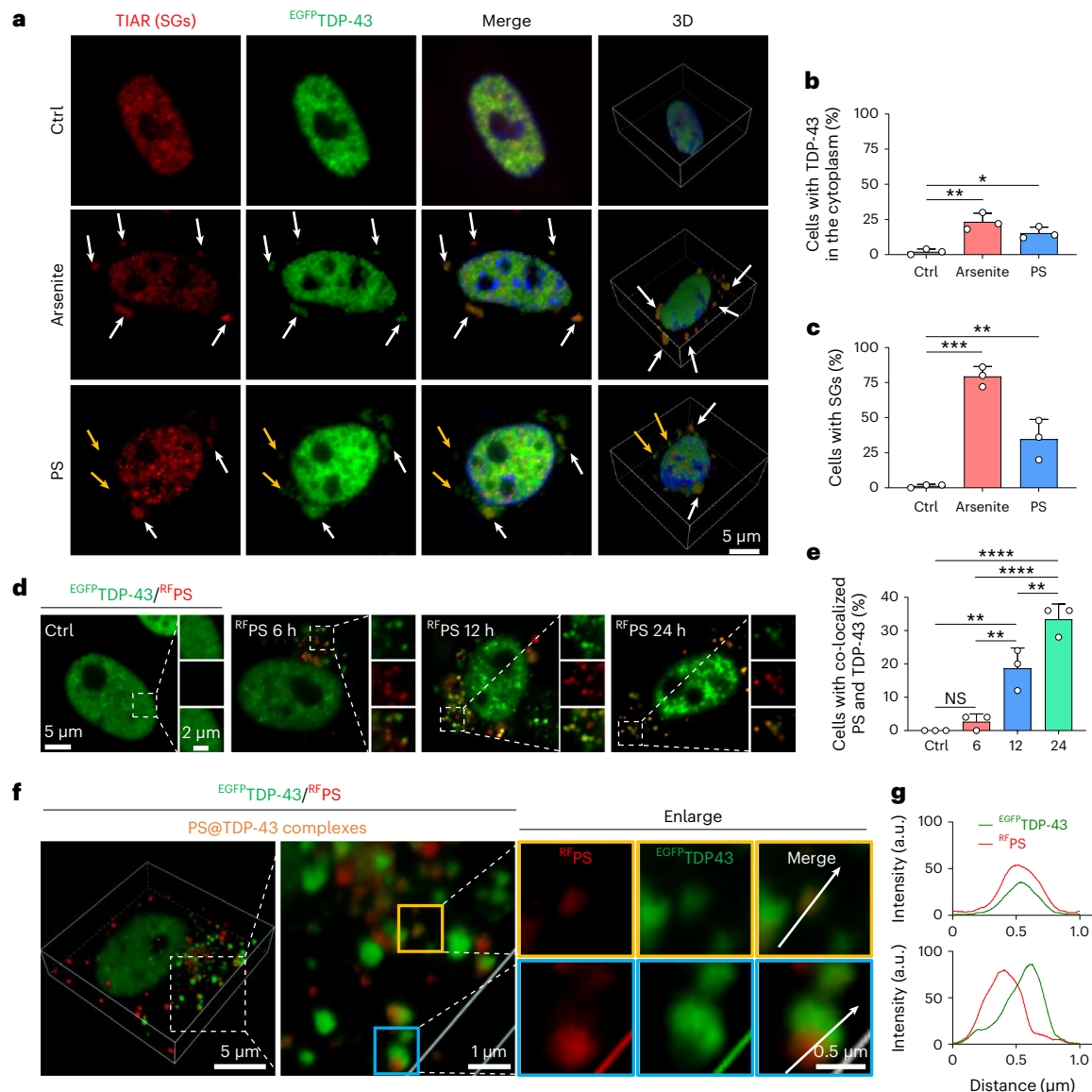


Fig. 3 | TDP-43 cytoplasmic translocation and the formation of PS@TDP-43 complexes. HeLa cells were treated with arsenite ($250 \mu\text{M}$, 30 min) or PS ($100 \mu\text{g ml}^{-1}$, 12 h). **a**, Cytoplasmic TDP-43 condensates co-localized with SGs are indicated by white arrows and cytoplasmic TDP-43 condensates isolated with SGs are indicated by yellow arrows. The rightmost column shows a three-dimensional (3D) reconstruction of z-stacks. DAPI (blue). **b**, The percentage of cells forming cytoplasmic TDP-43 condensates after arsenite or PS treatment. Data are expressed as mean $\pm$ s.d. ($n = 3$, 50 cells per experiment), as determined by one-way ANOVA with Tukey's multiple comparison tests (Ctrl versus arsenite, $P = 0.0025$; Ctrl versus PS, $P = 0.0237$). **c**, The percentage of cells forming SGs after arsenite or PS treatment. Data are expressed as mean $\pm$ s.d. ($n = 3$, 50 cells per experiment), as determined by one-way ANOVA with Tukey's multiple comparison tests (Ctrl

versus arsenite, $P = 0.0001$; Ctrl versus PS, $P = 0.0099$). **d**, The spatial relationship between internalized RFPS and cytoplasmic EGFP-TDP-43 condensates. Yellow, PS@TDP-43 complexes. **e**, The percentage of cells with co-localized EGFP-TDP-43 condensates and RFPS were determined. Data are expressed as mean $\pm$ s.d. ($n = 3$, 25 cells per experiment), as determined by one-way ANOVA with Tukey's multiple comparison tests (Ctrl versus PS 6 h, $P = 0.8452$; Ctrl versus PS 12 h, $P = 0.0020$; Ctrl versus PS 24 h, $P < 0.0001$; PS 6 h versus PS 12 h, $P = 0.0052$; PS 6 h versus PS 24 h, $P < 0.0001$; PS 12 h versus PS 24 h, $P = 0.0087$). **f**, Three-dimensional reconstruction images showed representative PS@TDP-43 complexes (yellow and blue boxes) in the cytoplasm. **g**, The fluorescence intensity profile along the indicated white line of PS@TDP-43 complexes in **f**. Top: encapsulation mode. Bottom: proximity mode. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$ (**b**, **c**, **e**).

exposed to cells, EGFP-TDP-43 NLS^{mut} accumulated in the cytoplasm. The dynamics of the PFPS@EGFP-TDP-43 NLS^{mut} were significantly lower than the EGFP-TDP-43 NLS^{mut} (Fig. 4e,f), suggesting a restrictive effect of PFPS on PFPS@EGFP-TDP-43 NLS^{mut} condensates recruited to SGs. As shown in Fig. 4g, arsenite-induced EGFP-TDP-43 NLS^{mut} condensates were co-localized with G3BP1 (a classical SG marker), which is consistent with previous studies^{16,18,38}. However, a portion of EGFP-TDP-43 NLS^{mut} is co-localized with PFPS (yellow arrow), rather than recruited to SGs. As the PFPS@EGFP-TDP-43 NLS^{mut} condensates lost their dynamics and were excluded from the 'safe compartment' SGs, they may develop into solid or fibril-like pathological structures.

As phosphorylation is one of the main hallmarks of TDP-43 proteinopathy³⁹, we explored the phosphorylation level of EGFP-TDP-43 NLS^{mut} condensates. The immunofluorescence assay showed that PFPS@EGFP-TDP-43 NLS^{mut} condensates (yellow box), instead of independent cytoplasmic EGFP-TDP-43 NLS^{mut} condensates (blue box), showed a remarkable pTDP-43 signal (Fig. 4h). Free PFPS (red box) was also found, which does not participate in forming pTDP-43 (Fig. 4h). The phosphorylation level of TDP-43 in the cytoplasm increased with the prolonged PS exposure (Supplementary Fig. 11). This result suggested that the sustained PS exposure not only provided continuous exogenous stress to the cells but also endowed sufficient time for PFPS@EGFP-TDP-43

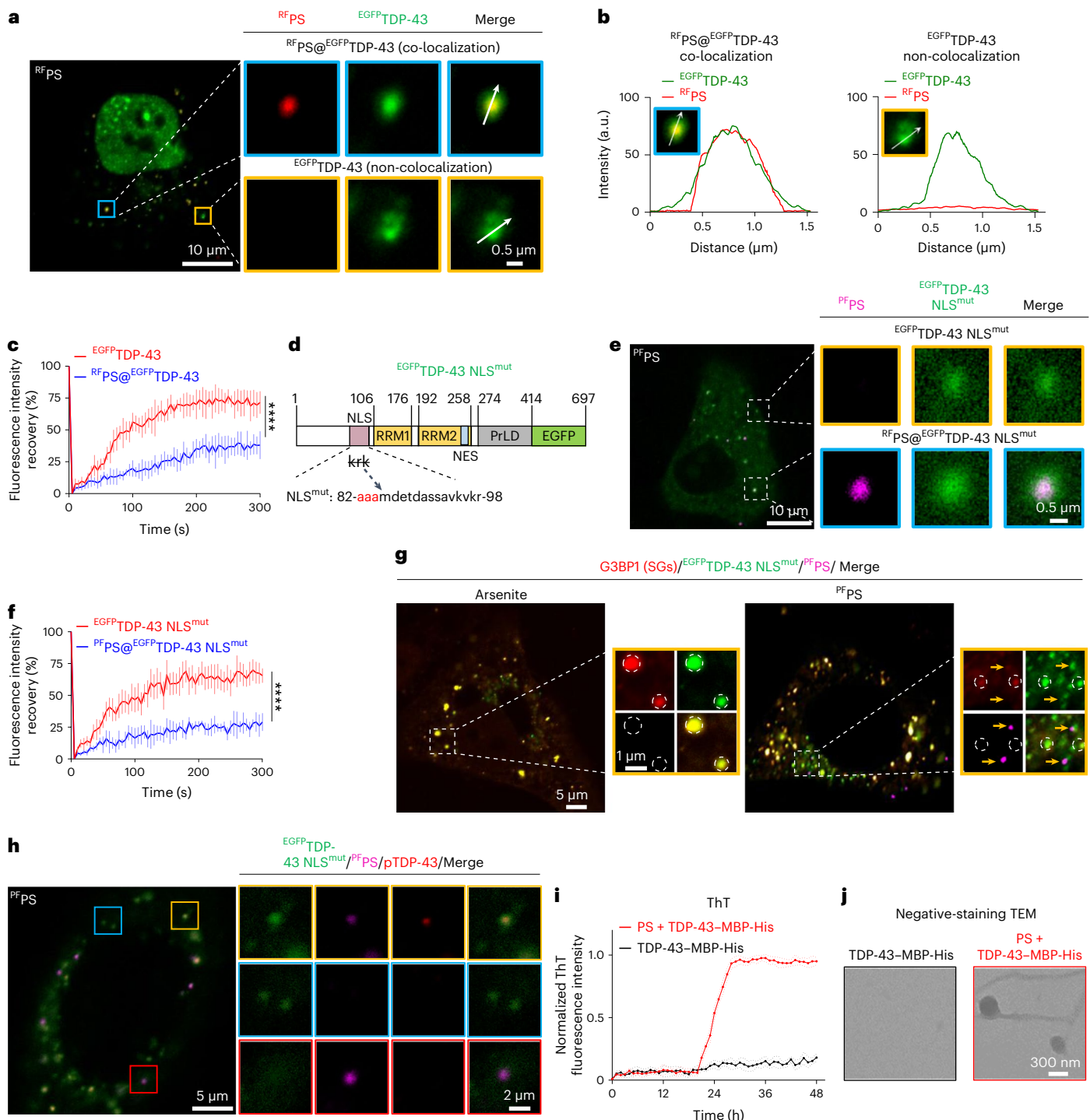


Fig. 4 | PS@TDP-43 complexes promote cytoplasmic TDP-43 condensation and solidification. HeLa cells were treated with RFPs (100 $\mu\text{g ml}^{-1}$) for 12 h. **a**, Representative images of RFPs@EGFP-TDP-43 condensates (blue box) and isolated EGFP-TDP-43 condensates (yellow box). **b**, The fluorescence intensity profile along the indicated white line of the condensates shown in **a**. **c**, FRAP recovery curves of RFPs@EGFP-TDP-43 and EGFP-TDP-43 condensates ($n = 6$ bleached condensates; graph represents mean $\pm$ s.d.). Statistical analysis was performed through one-sided two-way ANOVA with Tukey's multiple comparison tests (EGFP-TDP-43 versus RFPs@EGFP-TDP-43, $P < 0.0001$). **d**, A diagram of the residues mutated in human TDP-43. NLS, nuclear localization sequence; RRM1, RNA recognition motif 1; RRM2, RNA recognition motif 2; NES, nuclear export sequence; PrLD, praline-rich domain-like domain. The original amino acids (krk) in the NLS were mutated to alanine in this study. **e**, PFPS partially co-localized

with EGFP-TDP-43 NLS^{mut} condensates in cytoplasmic. **f**, FRAP recovery curves of RFPs@EGFP-TDP-43 NLS^{mut} and EGFP-TDP-43 NLS^{mut} condensates ($n = 6$ bleached condensates; graph represents mean $\pm$ s.d.). Statistical analysis was performed through one-sided two-way ANOVA with Tukey's multiple comparison tests (EGFP-TDP-43 NLS^{mut} versus RFPs@EGFP-TDP-43 NLS^{mut}, $P < 0.0001$). **g**, The localization of arsenite or PFPS-induced EGFP-TDP-43 NLS^{mut} condensates with SGs in the cytoplasm. **h**, The phosphorylated TDP-43 levels in RFPs@EGFP-TDP-43 NLS^{mut} and EGFP-TDP-43 NLS^{mut} condensates. Yellow box, RFPs@EGFP-TDP-43 NLS^{mut} condensates; blue box, EGFP-TDP-43 NLS^{mut} condensates; red box, PFPS. **i**, ThT fluorescence assay by co-incubation of TDP-43-MBP-His (10 μM) with PS (5 $\mu\text{g ml}^{-1}$). Data are expressed as mean $\pm$ s.d. ($n = 3$; dashed lines were plotted to indicate the s.d.). **j**, Negative-staining TEM images. TDP-43-MBP-His (10 μM) was incubated with PS (5 $\mu\text{g ml}^{-1}$) for 48 h. **** $P < 0.0001$ (**c**, **f**).

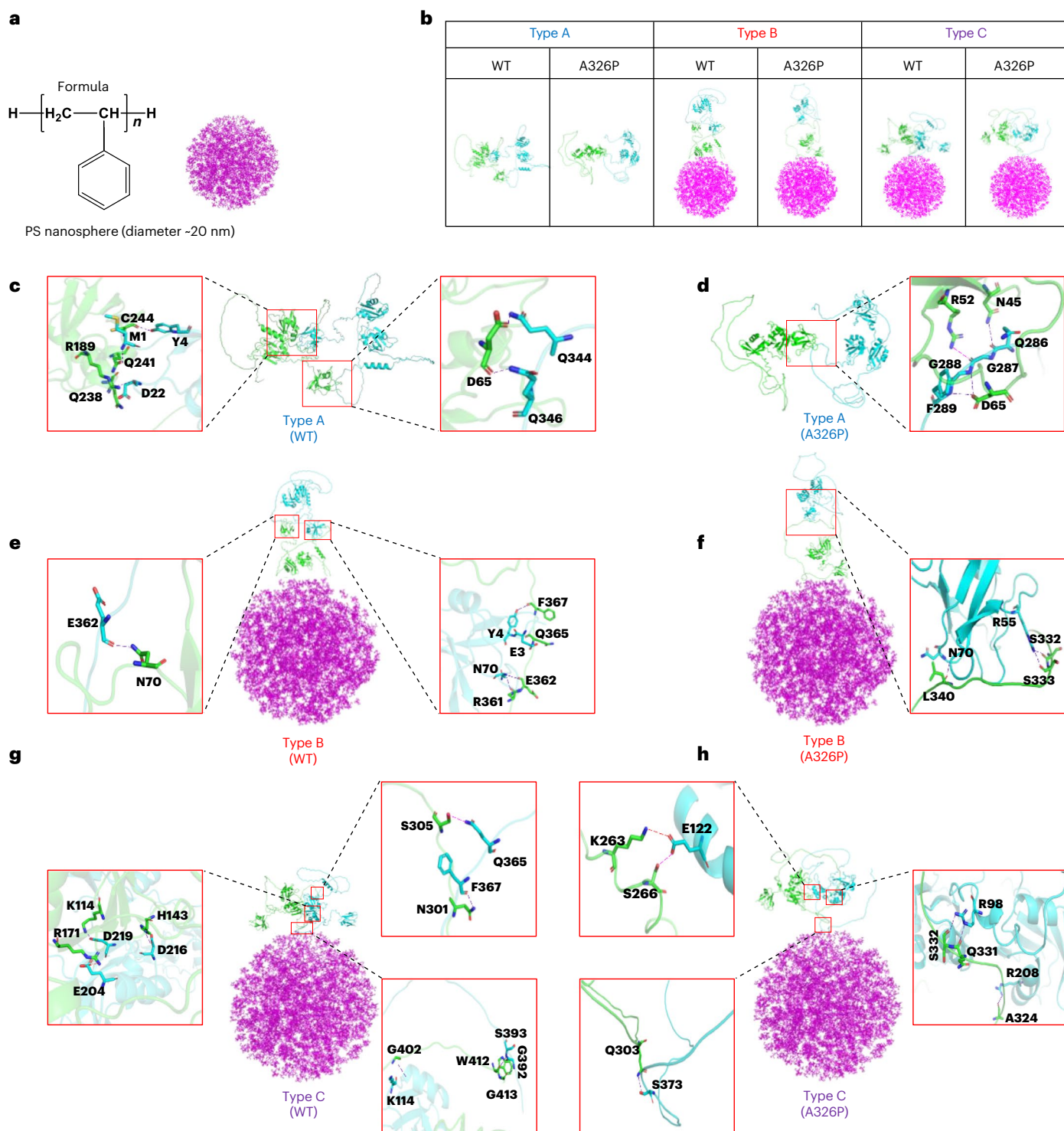


Fig. 5 | Molecular docking analysis of the effect of PS adsorption on the binding affinity of two TDP-43 molecules. a, PS nanosphere containing 1,400 PS monomers with ~20 nm in diameter. **b**, Schematic diagram of wild-type (WT) or A326P-mutated TDP-43 intermolecular docking in the presence or absence of PS. Type A, two TDP-43 molecules without PS. Type B, PS was placed adjacent to the TDP-43 dimer, while PS was associated with only one TDP-43 molecule.

Type C, PS was placed to bind with two TDP-43 molecules simultaneously. For the A326P mutation, the original alanine (326 site) of wild-type TDP-43 was mutated to proline. The structure of A326P-mutated TDP-43 was predicted by AlphaFold2. **c–h**, Representative three-dimensional schematics of TDP-43 intermolecular docking and corresponding residue interaction sites. **c**, Type A (WT). **d**, Type A (A326P). **e**, Type B (WT). **f**, Type B (A326P). **g**, Type C (WT). **h**, Type C (A326P).

formation, and both may be potential drivers of cytoplasmic TDP-43 phosphorylation. The conversion of $\text{PFPS@EGFP-TDP-43 NLS}^{\text{mut}}$ condensates into aggresomes was verified by an aggresomal marker, Proteo-Stat (Supplementary Fig. 12).

The formation of aggresomes is associated with pre-formed solid-like TDP-43 NLS^{mut} condensates containing a β -sheet structure,

which blocks dynamic exchange with soluble TDP-43 NLS^{mut} and facilitates TDP-43 NLS^{mut} aggregation²⁶. Furthermore, our data showed that PS promoted the formation of amyloid fibrils from TDP-43–MBP–His by co-incubating the recombinant TDP-43–MBP–His with PS and detected by thioflavin T (ThT) fluorescence assay (Fig. 4i) and negative-staining transmission electron microscopy (TEM) analysis (Fig. 4j).

Table 1 | Docking score of intermolecular TDP-43

Docking type	WT I_{sc} (REU)	A326P I_{sc} (REU)
Type A	−26.675	−22.128
Type B	−29.444	−25.505
Type C	−35.583	−28.469

I_{sc} , interface score, which represents the energy of the interactions across the interface. As the Rosetta energy function is a combination of physics- and statistics-based potentials, it does not attempt to match up with actual physical energy units, for example, kcal mol^{−1} or kJ mol^{−1}.

Molecular docking simulation of PS and TDP-43 interaction

The adsorption of proteins onto the surface of nanoparticles frequently leads to the denaturation of native protein structures, and previous studies have demonstrated that their interplay promotes the rate of protein fibrillation⁴⁰. Three molecular docking modalities were applied to assess the effect of PS adsorption on the binding affinity of two TDP-43 molecules (Fig. 5). A PS nanosphere with an approximate diameter of 20 nm consisting of 1,400 PS monomers was constructed (Fig. 5a; see Methods for details). The intermolecular docking for two TDP-43 molecules in the absence of PS was first presented (type A). As shown in Fig. 5b, TDP-43 molecules have binding sites and interactions in the low-complexity domain (residues 274–414), which was considered as the primary driving force for the TDP-43 LLPS¹⁷. The molecular docking results showed more residue interaction and lower docking scores between TDP-43 molecules associated with PS nanospheres (types B and C), compared with independent TDP-43 intermolecular docking (type A) (Fig. 5c–h and Table 1). Only one TDP-43 molecule was adsorbed on the surface of PS in type B, while two TDP-43 molecules were both adjacent to PS in type C. Compared with type B, type C has further elevated the intermolecular interactions and sites of TDP-43 dimer (Supplementary Fig. 13 and Supplementary Table 1). Indeed, the type C interface has the lowest docking score (−35.583 Rosetta energy units (REU)), followed by type B (−29.444 REU) and type A (−26.675 REU). The stability of the three interfaces is in good agreement with the number of contacts in the respective interfaces. In parallel, we utilized AlphaFold2 to predict the structure of the A326P-mutated TDP-43 (the original alanine at the 326 site of wild-type TDP-43 was mutated to proline) and simulated its interaction with PS. The interaction of the A326P TDP-43 dimer was relatively weaker than the wild-type TDP-43 dimer (−22.128 REU versus −26.675 REU). Consistently, with the presence of PS, the binding affinities of A326P TDP-43 dimer were lower than that of wild-type TDP-43 dimer in both type B and C styles, respectively (type B, −25.505 REU versus −29.444 REU; type C, −28.469 REU versus −35.583 REU). This finding describes the potential mechanism through which the formation of the PS@TDP-43 complexes accelerates TDP-43 LLPS and condensation.

PS exposure causes ALS-like neurodegeneration in mice

Accumulating epidemiological evidence supports the implication of airborne particulate exposure in ALS risk⁷. Thus, to confirm that chronic inhalation of PS leads to ALS-like syndrome in mice, an animal study was conducted correspondingly (Fig. 6a). To ensure that PS could breach the blood–brain barrier and overcome the organism's clearance to generate remarkable brain damage, C57BL/6J mice were intranasally instilled with 30 nm PS at 20 mg kg^{−1} d^{−1} for 28 continuous days⁵. Nanoparticles within this size range can pierce into the brain through the nasal/olfactory nerves and systemic circulation. We first characterized the pathological features of ALS. Although mice showed significant weight loss, muscle fibres from the biceps did not show noticeable muscle wasting compared with the Ctrl group (Supplementary Fig. 14a–c). Meanwhile, there was no significant degeneration or

loss of lower motor neurons from the anterior horn of the spinal cord compared with the Ctrl group (Supplementary Fig. 14d,e). In the early stages, ALS disease primarily manifests as movement impairment, and as the disease progresses, muscle wasting and weakness gradually become apparent. We then verified that PS intranasal exposure caused abnormal behaviours in mice. As shown, compared with the control animal, PS-exposed mice showed apparent recession in activity (Fig. 6b and Supplementary Fig. 15), motor and coordination (Fig. 6c), grasp and manoeuvrability (Fig. 6d), limb grip strength (Fig. 6e), and novel object recognition and cognitive abilities (Fig. 6f,g). Gene set enrichment analysis (GSEA) on the mouse brains' transcriptomic data supported these findings. It showed significant down-regulation of ALS pathways in the mouse brains with PS exposure, implying the induction or facilitation of ALS development (Fig. 6h). Further differential expression gene analysis indicated that mitochondrial DNA, ATP synthase FO subunit 8 (ATP8) was dysregulated. The protein encoded by the *ATP8* gene plays a vital role in the energy production of mitochondria, and mitochondrial dysfunction is postulated to be central to ALS⁴¹.

ALS is an irreversible and lethal motor neuron disease characterized by degeneration of upper motor neurons and lower motor neurons. Damage to upper motor neurons leads to uncontrolled movement, decreased sensitivity to superficial reflex stimulation and spasticity⁴². On the contrary, lower motor neuron damage is usually associated with muscle atrophy, weakness and hyporeflexia without sensory involvement. Primarily, the cortex, brainstem and spinal cord motor neurons suffer degeneration⁴³. Given previous reports of nanoparticle deposition in the brain, we focus primarily on the related pathologic features in the cerebral cortex. As the loss of neurons is one of the crucial hallmarks of ALS, we used the Nissl staining assay to observe the aberrant neurons caused by PS. Indeed, PS intranasal exposure significantly diminished the number of Nissl bodies in nerve cells in the cortex region compared with the control (Fig. 6i,j). The deposition of abnormal hyperphosphorylated TDP-43 was previously identified in the nerve bundles of patients with ALS⁴⁴. To this end, an immunofluorescence assay was used to detect pTDP-43 at S409/410 in mouse brain tissue. As shown in Fig. 6k,l, pTDP-43 foci were markedly formed in the PS group. In addition, TDP-43 granules in cortical regions were found to be AmyT positive in the PS-exposed group (Supplementary Fig. 16). TDP-43 formed condensates in the brain of PS-treated mice, both in the cytoplasm and the nucleus (Fig. 6m, PS group, red box, yellow arrows).

Conclusion

Our present research connects environmental pollutants with protein LLPS and proposes a specific mechanism linking PS exposure to TDP-43 proteinopathy. This exposure leads to abnormal LLPS of nuclear TDP-43 through oxidative stress and the depletion of Hsp70, furthering the liquid-to-solid phase transition and pathological clustering of TDP-43. In addition, creating the PS@TDP-43 complex amplifies the likelihood of TDP-43 amyloid fibrillation (Extended Data Fig. 5). Notably, although PS exposure leads to phenotypes associated with ALS, it also links with widespread stress responses, including neuro-inflammation and oxidative stress responses. Currently, it is unclear whether PS exposure leads to broad toxicity of aggregation-prone proteins (for example, α -synuclein or tau, which are involved in Parkinson's disease and Alzheimer's disease, respectively). The observed brain pathology and dysfunction suggests that cognitive changes explain the 'ALS-like' phenotypes; however, the lack of motor neuron degeneration and muscle wasting indicate the difference in PS exposure with genetic/inherited ALS. In fact, TDP-43 pathology remains rather sparse in the patient's brains, contrary to what we observed in these mice. These discoveries contribute fresh evidence and perspectives for evaluating how ultrafine particles may initiate or intensify NDDs.

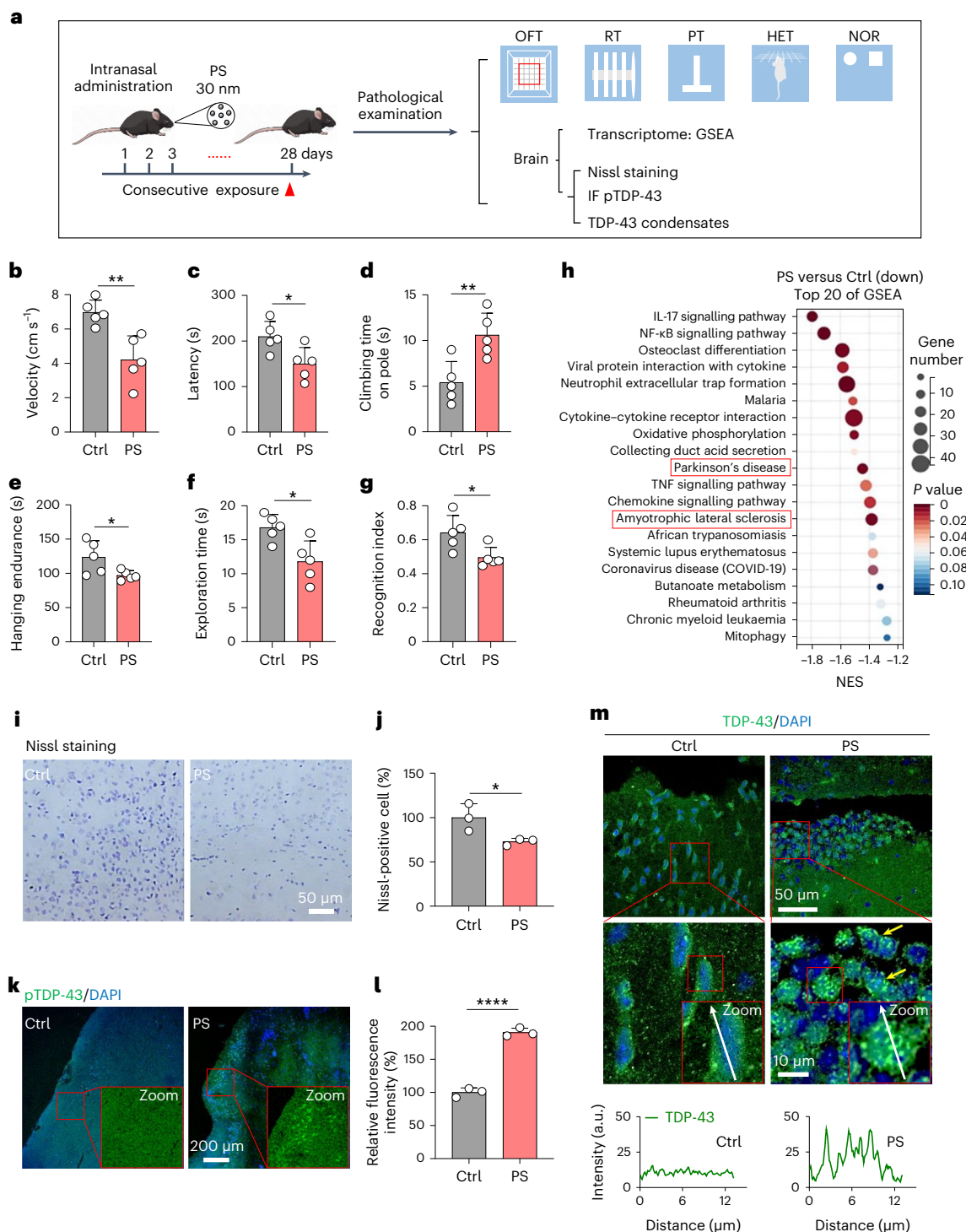


Fig. 6 | PS intranasal exposure causes ALS-like neurodegeneration and liquid droplet-like TDP-43 formation in mice. **a**, Flow chart of PS intranasal exposure in a mouse model. OFT, open-field test; RT, rotarod test; PT, pole test; HET, hanging endurance test; NOR, novel object recognition; IF, immunofluorescence. **b**, Velocity of movement in the open-field test. **c**, Latency time in the rotarod test. **d**, Climbing time in the pole test. **e**, Hanging endurance time in the hanging endurance test. **f**, The time of exploration for novel objects. **g**, Recognition index in the novel object recognition test. Data are expressed as mean $\pm$ s.d., $n = 5$ independent mice per group in **b–g**, as determined by an unpaired two-tailed t -test. $P = 0.0044$ (**b**); $P = 0.0245$ (**c**); $P = 0.0082$ (**d**); $P = 0.0427$ (**e**); $P = 0.0144$ (**f**); $P = 0.0248$ (**g**). **h**, GSEA on the mouse brains' transcriptome ($n = 3$). The GSEA

result is based on a one-sided Kolmogorov–Smirnov test. **i**, Representative images of Nissl staining of the brain's cortex region in mice. **j**, Relative percentage of positive-Nissl-staining cells in **i**. Data are expressed as mean $\pm$ s.d. ($n = 3$), as determined by an unpaired two-tailed t -test ($P = 0.0421$). **k**, Representative immunofluorescence images of pTDP-43 in the brain's cortex region. **l**, The percentage of pTDP-43 fluorescence intensity in **k**. Data are expressed as mean $\pm$ s.d. ($n = 3$), as determined by an unpaired two-tailed t -test ($P < 0.0001$). **m**, Representative immunofluorescence images and fluorescence intensity profile along the indicated white line of TDP-43 condensates in the brain's cortex region. $*P < 0.05$, $**P < 0.01$, $****P < 0.0001$ (**b–g, j, l**).

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41565-024-01683-5>.

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Methods

PS characterization

PS (30 nm, fluorescence free, Zhongkeleiming Science and Technology), PS (300 nm, fluorescence free, BaseLine), ^{RF}PS (300 nm, $E_x = 540$ nm, $E_m = 580$ nm, BaseLine) and ^{PF}PS (300 nm diameter, $E_x = 620$ nm, $E_m = 680$ nm, BaseLine). TEM analysis was conducted with a JEM 1200EX (JEOL). The hydrodynamic diameters and zeta potential of PS were determined by dynamic light scattering with Zetasizer Nano-ZS ZEN 3600 (Malvern Instruments) and analysed with Zeta View 8.05.04.

Plasmid construction

pcDNA3.1-TDP-43 and pCAG-^{EGFP}TDP-43 plasmids were obtained as a generous gift from Prof. Yanshan Fang. The pcDNA3.1-TDP-43-A326P-hemagglutinin (HA) plasmids were generated by site-directed mutagenesis using the Fast Mutagenesis Kit II (Vazyme). To generate the pCAG-^{RFP}Hsp70 plasmids, the Hsp70 and red fluorescent protein (RFP) coding sequences were amplified from pET28a-His-Hsp70 and pCAG-RFP, respectively, then subcloned into pCAG plasmid using the ClonExpress MultiS One Step Cloning Kit (Vazyme)²⁴. pcDNA3.1-TDP-43 was used as the template to generate the following plasmid. pcDNA3.1-^{EGFP}TDP-43 NLS^{mut} plasmid was generated by site-directed mutagenesis using the Fast Mutagenesis Kit MultiS (Vazyme). For *Escherichia coli* expression, the gene of TDP-43 was cloned into pET9d with a tobacco etch virus (TEV) protease cleavable MBP-His tag at the C-terminus (pET9d-TDP-43-TEV-MBP-His as a generous gift from Prof. Yanshan Fang).

In vitro protein expression and purification

TDP-43-MBP protein was expressed in BL21(DE3) pLysS *E. coli* cells (TransGen Biotech) as previously described²⁴. Briefly, cells were induced to overexpress the protein using 1 mM isopropyl β -D-1-thiogalactopyranoside (Sigma-Aldrich) at 16 °C for 16 h. Cells were collected and lysed by lysis buffer (50 mM Tris-HCl, pH 7.5, 1 M NaCl, 2 mM DTT, 10% glycerol, 1 mM EDTA and 2 mM PMSF). After collection by centrifugation, the protein was loaded onto a MBP-Sep Dextrin 6FF chromatography column (Yeasen Biotechnology) and then eluted with elution buffer (50 mM Tris-HCl, pH 7.5, 1 M NaCl, 2 mM DTT, 10% glycerol and 10 mM maltose). Eluted protein was further purified using gel filtration chromatography (Superdex 200 16/300; GE Healthcare) equilibrated with storage buffer (50 mM Tris-HCl, pH 7.5, 300 mM NaCl and 2 mM DTT). The protein was concentrated, flash frozen in liquid nitrogen and stored as aliquots at -80 °C until use.

In vitro recombinant protein LLPS assay

Full-length TDP-43-MBP (30 μ M or 40 μ M) was mixed with 30 nm PS (0, 0.5 μ g ml⁻¹, 5 μ g ml⁻¹ and 50 μ g ml⁻¹) with (or without) 15% dextran 70 in 50 mM Tris-HCl, 100 mM NaCl, 15% dextran 70, pH 7.5.

Cell culture and transfection

HeLa cells (Procell Life Science and Technology, CL-0101) were cultured in Dulbecco's modified Eagle's medium (DMEM, KeyGen BioTECH) supplemented with 10% (v/v) fetal bovine serum (FBS, Gibco) and 1% penicillin/streptomycin at 37 °C in a humidified incubator containing 5% CO₂. Lipofectamine 2000 plus transfection reagent (US Everbright) was used to transfect plasmids into HeLa cells. Cells were transfected for at least 24 h before the subsequent PS exposure or other treatment. For the knockdown experiment, siRNA (Genepharma, A09001) was transfected into HeLa cells using the Lipofectamine 2000 Reagent (Invitrogen) according to the manufacturer's instructions. siRNA was incubated for 72 h before cells were collected. siRNA sequences: si-Ctrl: 5'-UUC UCC GAA CGU GUC ACG UTT-3' and si-HSPA1A: 5'-CCA AGC AGA CGC AGA UCU UTT-3'. The primary and secondary antibodies used in this study are listed in Supplementary Table 2.

PS and arsenite treatment

HeLa cells were seeded in confocal dishes (35 mm) overnight at a density of 1×10^6 cells per dish and then transfected with indicated plasmids or siRNA. Cells were treated with 250 μ M arsenite (NaAsO₂) for 30 min or 25 μ g ml⁻¹, 50 μ g ml⁻¹ and 100 μ g ml⁻¹ PS for 6 h, 12 h and 24 h. Subsequent experiments were performed after the indicated treatment time.

Confocal imaging

Fluorescence imaging was performed on a confocal fluorescence microscope (Nikon, N-SIM E) using a $\times 60$ oil immersion objective (numerical aperture 1.4). Confocal images were then processed in NIS-Elements AR 5.01.00 64-bit (Nikon).

Three-dimensional rendering

HeLa cell nuclei were scanned from top to bottom to capture -60 images (0.45 μ m per stack) using the Nikon confocal microscope z-axis scanning system, and the maximum intensity projection was used to construct the final image series.

Fusion event imaging

HeLa cells (1×10^6 cells per dish) transfected with pCAG-EGFP-TDP-43 were grown on confocal dishes (35 mm) for 24 h before PS treatment. Fusion event imaging was performed on a confocal fluorescence microscope (ND acquisition-delayed capture module). To view the fusion behaviour of TDP-43 condensates, images were collected at 10 s intervals.

FRAP

Intracellular ^{EGFP}TDP-43 condensates were bleached in unfixed HeLa cells with a 488 nm laser at 80% intensity for 3 s. The bleached area was artificially focused on a circular 2- μ m-diameter region of interest. After photobleaching, time-lapse images were captured every 5 s for 5 min. For each indicated time point (t), the fluorescence intensity within the bleached droplet was normalized to the fluorescence intensity of a nearby unbleached droplet. The normalized fluorescence intensity of pre-bleaching was set to 100%, and the normalized fluorescence intensity at each time point (I_t) was used to calculate the fluorescence recovery (FR) according to the following formula: $FR(t) = (I_t/I_{pre-bleaching}) \times 100\%$ (ref. 38). ImageJ 1.49p and GraphPad Prism 8.0.2 were used to measure and analyse the FRAP data.

ROS level

HeLa cells (1×10^6 cells per dish) were grown on confocal dishes (35 mm) and exposed to NAC (2 mM) and/or PS (100 μ g ml⁻¹) for 6 h. Cells in the control group received serum-free medium treatment for 6 h and were then incubated with DCFH-DA (Sigma-Aldrich) probe in the dark at 37 °C for 30 min. Cells were washed three times with room-temperature PBS. Data were collected using a BD Melody FACS flow cytometer with BD FACShorus 1.3 software and analysed using FlowJo v.10 software.

MMP assay

HeLa cells (1×10^6 cells per dish) were grown on confocal dishes (35 mm) and then incubated overnight. Fresh DMEM media with PS (100 μ g ml⁻¹) was added to the confocal dishes. Mitochondrial membrane potential (MMP) was determined using the JC-1 MMP assay kit (Yeasen Biotechnology) according to the manufacturer's instructions.

Lysosomal damage

HeLa cells (1×10^6 cells per dish) were grown on confocal dishes (35 mm) and then incubated overnight. After treatment, cells were stained with acridine orange (5 μ g ml⁻¹) (Amresco) at 37 °C for 30 min for assessment of lysosomal permeability. Fluorescence images and signals were detected by confocal microscopy and flow cytometer, respectively.

Cathepsin B release

HeLa cells (1×10^6 cells per dish) were grown on confocal dishes (35 mm) and then incubated overnight. PS-challenged cells were stained with Magic Red (ImmunoChemistry Technologies) solution (200 μ l) at 37 °C for 30 min. Then nuclei were stained with Hoechst 33342 (Yeasen Biotechnology) for 10 min at room temperature. Finally, the fluorescence images and signals were detected by confocal microscopy and flow cytometer, respectively.

Cell membrane damage

HeLa cells (1×10^6 cells per well) were grown on 6-well plates and then incubated overnight. After treatment, cells were stained with propidium iodide (2 μ M) (Yeasen Biotechnology) at 37 °C for 30 min for analysis of cell membrane integrity. Fluorescence signals were detected by flow cytometer.

^{RF}PS internalization

HeLa cells (1×10^6 cells per dish) were seeded in 6-well plates and incubated with a serum-free medium containing ^{RF}PS. Then the cells were washed three times with PBS and collected by centrifugation. The fluorescence signals of ^{RF}PS were detected by flow cytometry.

1,6-HD treatment

PS-containing medium was replaced with fresh medium containing 3% (v/v) 1,6-HD (Sigma). The condensate disappearance process was observed using a confocal fluorescence microscope. Subsequently, the 1,6-HD-containing medium was replaced with the 1,6-HD-free medium at the ninth minute, and the condensation recovery process was observed.

Immunocytochemistry

HeLa cells (1×10^6 cells per dish) were grown on confocal dishes (35 mm) and then incubated overnight. Fresh DMEM media with indicated concentrations of PS was introduced to the confocal dishes. Cells were then fixed with 4% paraformaldehyde at room temperature for 15 min and washed 3 times with PBS. Cells were permeabilized and blocked with blocking buffer (1% Triton X-100, 3% BSA) for 2 h. Cells were then incubated with corresponding primary antibodies overnight at 4 °C. After washing 3 times with PBS, cells were incubated with corresponding fluorescently labelled secondary antibodies diluted at 1:500 in PBS for 1 h at room temperature. The nucleus was stained with DAPI for 10 min.

Protein extraction and western blotting

The total protein of cultured cells was lysed with 4% extraction buffer (100 mM Tris pH 6.8, 4% SDS, 1% mercaptoethanol, 40% glycerol and 0.04% bromophenol blue) containing protease inhibitor cocktail and phosphatase inhibitor (Beyotime Biotechnology). Nuclear/cytoplasmic subcellular fractionation was performed using a Nuclear and Cytoplasmic Protein Extraction Kit (Beyotime Biotechnology) according to the manufacturer's instructions. The protein samples were then boiled at 100 °C for 10 min and separated by 10% or 12.5% SDS-PAGE, transferred to nitrocellulose filter membrane and blocked with 5% skimmed milk in Tris-buffered saline with Tween 20 (TBST). Then membranes were probed with primary antibodies at 4 °C overnight. After washing 3 times with TBST solution, the bound primary antibodies were probed with 1:2,000 dilution of HRP-conjugated affiniPure goat anti-rabbit IgG(H+L) and then the protein signals were visualized with BeyoECL Star analysis reagent (Beyotime Biotechnology).

Amyloid and ProteoStat staining

The amyloid dye AmyT (Ebba Biotech) was used essentially as described in the manufacturer's guidelines. Cells were fixed with 4% paraformaldehyde for 10 min, washed 3 times with PBS and then permeabilized with Triton X-100 0.2% (v/v) for 5 min, followed by 3 washes with PBS. AmyT with a dilution of 1:500 was introduced and incubated for 30 min

at room temperature. Cells were washed three times and counter-stained with DAPI. For ProteoStat staining, fixed cells were bathed with ProteoStat dye (Enzo Life Sciences) at room temperature for 30 min. After washing three times with PBS, cells were stained with DAPI for 10 min, and fluorescence imaging was performed using the confocal fluorescence microscope.

Cell viability

Cell viability was detected using Cell Counting Kit-8 (CCK-8) (Bimake), according to the manufacturer's instructions. Briefly, 10 μ l of CCK-8 solution was added to each well and incubated at 37 °C for 2 h. The absorbance of each culture well was measured with a microplate reader (BioTek ELX-800) at 450 nm.

Redox western blotting

Redox western blotting analysis for Hsp70 was performed according to a previously reported method⁴⁵. The total protein of cultured cell was lysed with lysis buffer (50 mM Tris, pH 7.4, 150 mM NaCl, 1% NP-40, 0.25% sodium deoxycholate, 1 mM PMSF, 1 mM EDTA, phosphatase inhibitors, sodium fluoride and leupeptin protease inhibitor). Then lysate samples were centrifuged at 12,000g at 4 °C for 10 min. The supernatant was collected as the whole cell extract. The samples were diluted in sample buffer with or without β ME. After heat denaturation, proteins were separated by 10% SDS-PAGE, and Hsp70 was visualized by western blotting.

TDP-43 adsorption

PS was introduced into HeLa cell lysate (or purified TDP-43-MBP-His solution) and incubated with a gentle shake at 37 °C for 1 h. The mixtures were centrifuged at 12,000g for 30 min, and the supernatant was discarded. Precipitation was washed three times with PBS and collected by centrifugation. Then each sample was denatured using a loading buffer and heated to 97 °C for 10 min. The amount of TDP-43 adsorbed on the surface of PS was semi-quantified by western blotting.

Formation of TDP-43 fibrils

For the ThT assay, 10 μ M of TDP-43-MBP-His sample solution (20 mM MES, pH 6.0, 150 mM NaCl, 4 mM DTT and 0.05% NaN₃) with or without the presence of PS (5 μ g ml⁻¹) was performed for ThT fluorescence kinetic detection. The samples were incubated with 20 μ M of ThT. The ThT fluorescence (excitation at 452 nm and emission at 48 nm) was monitored using a SpectraMax iD3 microplate reader (Molecular Devices) at 37 °C. The morphology of TDP-43-MBP-His fibrils was visualized by TEM.

Molecular docking

The structure of TDP-43 was obtained from AlphaFoldDB (AF-Q13148-F1). In parallel, the structure of A326P-mutated TDP-43 (the original alanine at 326 site of wild-type TDP-43 was mutated to proline) was predicted by AlphaFold2. PACKMOL 18.169 software was used to construct the nanosphere containing 1,400 PS monomers with ~20 nm diameter. The global docking module in ROSETA2020 was used to carry out molecular docking and analyse residue docking sites and interactions. The energy of interaction between two TDP-43 molecules was assessed by the interface interaction score. Three kinds of docking modality were simulated. First, molecular docking was applied to elucidate the binding of two TDP-43 molecules without PS (type A). Next, PS was placed adjacent to the TDP-43 dimer, while PS was associated with only one TDP-43 molecule (type B). Third, PS was placed to bind with two TDP-43 molecules simultaneously (type C). The binding energy values in arbitrary REU (Rosetta) were calculated, and the corresponding binding sites were predicted. Computational molecular docking graphs were performed with Pymol 2.5.7.

Animal care

Male C57BL/6J mice (7–8 weeks old) were purchased from Hunan SJA Laboratory Animal. All animals were supplied with free access to food and water and maintained in a 23–25 °C and 50–60% humidity environment with a 12 h/12 h light/dark cycle. All animal studies were approved by the Institutional Animal Care and Use Committee of Southwest University (IACUC-20230313-04) and followed the NIH's guidelines for the care and use of laboratory animals.

Intranasal instillation

After the acclimation period, mice were randomly divided into the control and PS group ($n = 5$ in each group). After anaesthesia with pentobarbital, mice were intranasally instilled with PS suspension (20 mg per kg body weight, every day for 28 consecutive days⁵), and the control group received the same volume of distilled water. The body weight of mice was daily recorded during the whole experiment. At the end of treatment, behavioural tests were conducted. Finally, mice were killed and perfused with ice-cold PBS, and the mice brain was collected for subsequent pathological or biochemical experiments.

Behavioural tests

Several behavioural tests, including open-field test, rotarod test, pole test, hanging endurance test and novel object recognition test, were conducted according to our previous publication⁴⁶.

RNA sequencing

Total RNA was extracted with TRIzol reagent (Invitrogen) according to the manufacturer's protocol. Total RNA quality was assessed using a NanoDrop 2000 spectrophotometer (Thermo Scientific). RNA integrity was evaluated using the Agilent 2100 Bioanalyzer (Agilent Technologies). Then the libraries were constructed using the VAHTS Universal V6 RNA-seq Library Prep Kit according to the manufacturer's instructions. The libraries were sequenced on a Illumina Novaseq 6000 platform, and 150-bp paired-end reads were generated. Raw reads of FASTQ format were first processed using fastp, and the low-quality reads were removed to obtain the clean reads. Clean reads were mapped to the reference genome using HISAT2. The fragments per kilobase of transcript per million mapped reads (FPKM) of each gene was calculated, and the reads of gene were obtained by HTSeq-count. Principal component analysis was conducted using R (v 3.2.0) to assess the biological duplication of samples.

GSEA

GSEA analysis was performed using GSEA software⁴⁷. The analysis used a predefined gene set, and the genes were ranked according to the degree of differential expression in the two types of sample.

Haematoxylin-and-eosin staining

The right biceps brachialis muscles of mice were fixed with 4% paraformaldehyde and then sliced into ~5- μ m-thick sections. The muscle tissue sections were then stained with haematoxylin-and-eosin staining. All the steps we performed above were in accordance with the standard procedure⁴⁸. Finally, muscle tissue sections were visualized and imaged under a microscope (OLYMPUS IX71) with cellSens Standard 1.2.1 (build 7542) software. The average cross-section of muscle fibres was obtained from 3 independent samples and 12 muscle fibres were randomly selected for each sample.

Nissl staining

Mice brains or spinal cord tissue were fixed in 4% paraformaldehyde overnight before paraffin embedding. The tissues were sectioned into ~5- μ m-thick slices. Then paraffin sections were dewaxed and stained with Nissl staining solution (Yeasen Biotechnology) for 10 min. Finally, tissue sections were visualized and imaged under a microscope (OLYMPUS IX71) with cellSens Standard 1.2.1 (build 7542) software.

and counted for Nissl-positive neurons in the cortex regions or anterior horn of the spinal cord. The signal intensity was quantified using ImageJ 1.49p software.

Immunofluorescence

Mouse brain tissue was dewaxed, sectioned and subsequently blocked with 3% BSA (Biosharp) for 30 min at room temperature. Then brain tissue sections were incubated with pTDP-43 (Ser409/410) and TDP-43 antibodies overnight at 4 °C, respectively. After washing 3 times with PBS, tissue sections were incubated with the corresponding fluorescently labelled secondary antibody (CoraLite488, Proteintech) diluted at 1:300 in PBS for 1 h at room temperature. Then tissue sections were stained with DAPI for 10 min and washed 3 times with PBS. Finally, tissue sections were observed and imaged using the confocal fluorescence microscope. The pTDP-43 fluorescence intensity was quantified using ImageJ 1.49p software.

Statistics and reproducibility

The data graphing and statistical analyses were performed with GraphPad Prism 8.0.2 (GraphPad Software). Data were generated from at least three independent experiments with similar results. Data were expressed as mean $\pm$ s.d., analysed via one-way or two-way analysis of variance (ANOVA), and followed by Tukey's multiple comparison test or Bonferroni's multiple comparison test as indicated in respective figure captions or two-tailed unpaired *t*-test. $P < 0.05$ was considered statistically significant.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The source data underlying the main text figures (Figs. 1–4 and 6 and Extended Data Figs. 1–4) and the supplementary figures (Supplementary Figs. 1, 4, 6, 7, 8, 10, 11, 14, 15) and videos (Supplementary Videos 1–8) are provided as source data files or deposited in Gene Expression Omnibus (RNA-seq) under accession number [GSE264018](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE264018). The description data for the TDP-43 structure were acquired from the AlphaFold Protein Structure Database (AlphaFoldDB) under the accession code [AF-Q13148-F1](https://alphafold.com/entry/Q13148-F1). TDP-43 protein disorder region scores from Predictor of Natural Disorder Regions (PONDR; an online tool, <http://www.pondr.com/>).

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Author contributions

H.S. and Y.S. conceived the study. H.S., B.Y., Q.L. and X.Z. performed assays and contributed to data interpretation. E.S. and Y.S. supervised the experiments. H.S., C.L. and Y.S. wrote the paper. E.S., Y.S. and G.J. provided reagents and oversaw the research. All authors had input on the content of the paper and approved the final version of the paper.

Competing interests

The authors declare no competing interests.

Additional information

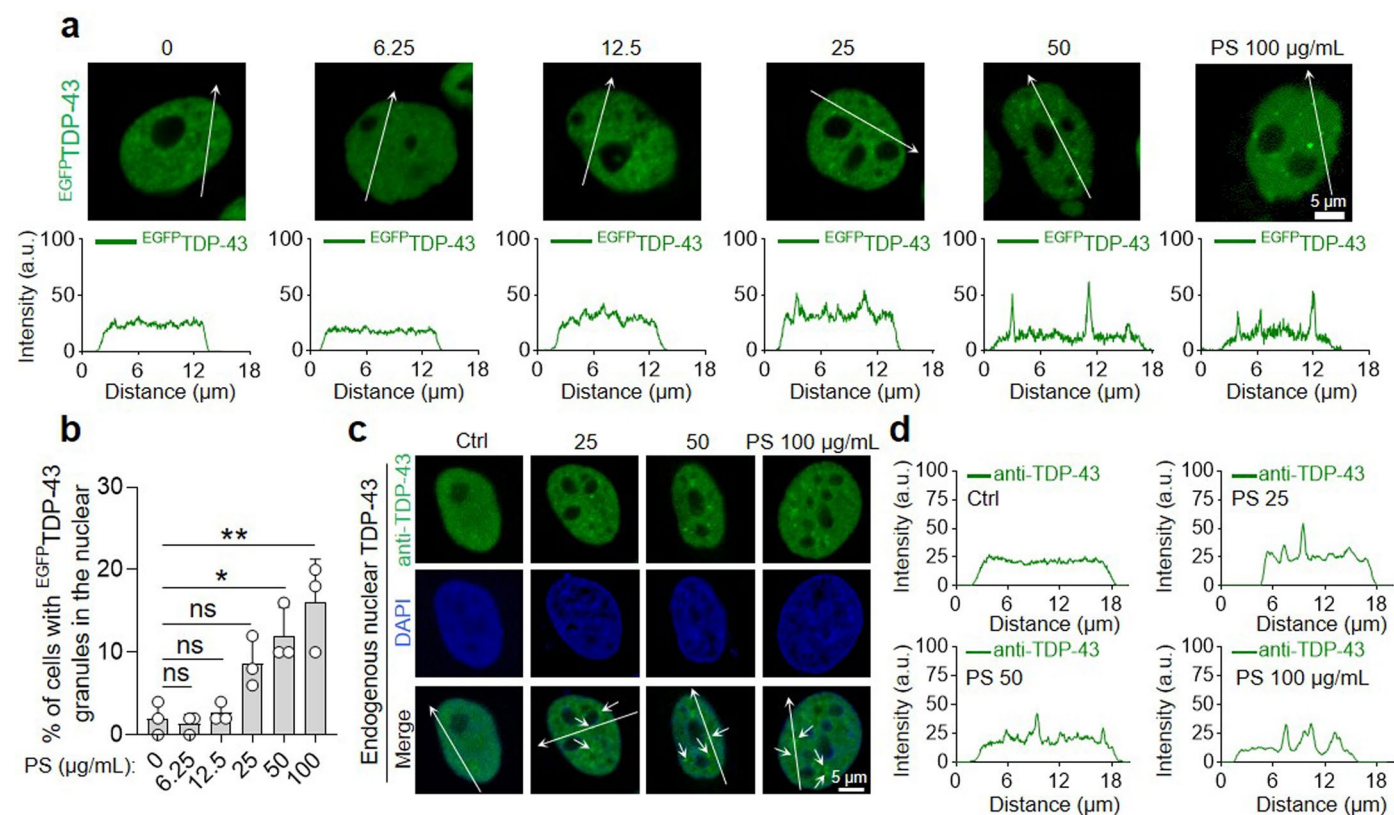
Extended data is available for this paper at <https://doi.org/10.1038/s41565-024-01683-5>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41565-024-01683-5>.

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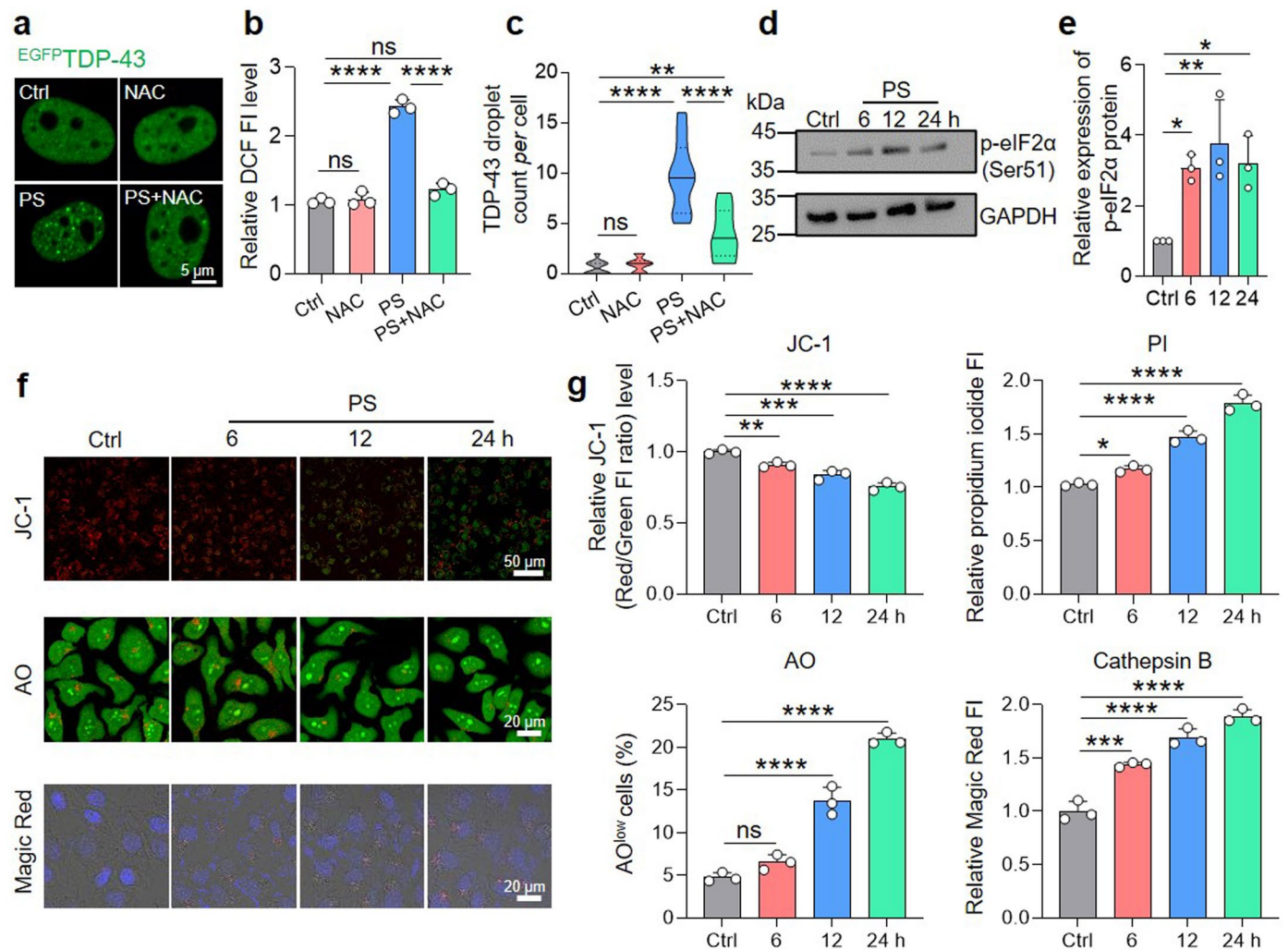
Peer review information *Nature Nanotechnology* thanks Pulong Li and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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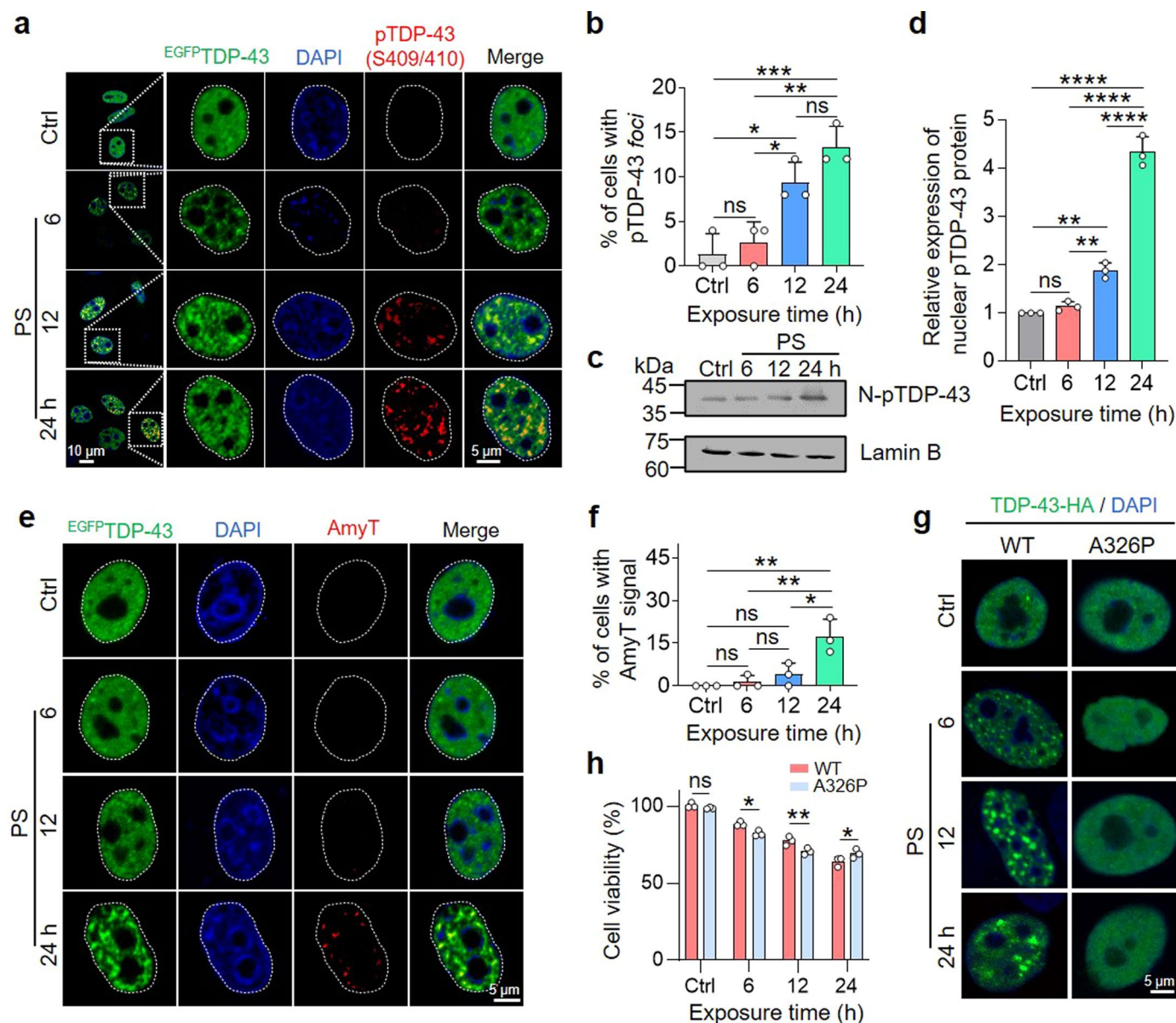
Extended Data Fig. 1 | PS exposure induces transgenic/endogenous nuclear TDP-43 granule formation in HeLa cells. a, Representative images and the fluorescence intensity profile along the indicated white line of HeLa cells forming ^{EGFP}TDP-43 granules at 6 h after the indicated concentrations of PS treatment. **b**, The percentage of cells forming ^{EGFP}TDP-43 granules in (a). Data are expressed as means ± SD. *n* = 3, 50 cells *per* experiment, as determined by one-way ANOVA

with Tukey's multiple comparison tests (*p* = 0.0167 and 0.0012 for PS of 50 and 100 µg/mL *vs.* 0 µg/mL, respectively; *ns*, *p* > 0.05). **c**, Representative images and **d**, fluorescence intensity profile along the indicated white line of HeLa cells forming endogenous TDP-43 granules (arrows) in response to indicated PS exposure concentrations after 6 h. *n* = 3 independent experiments *per* group.



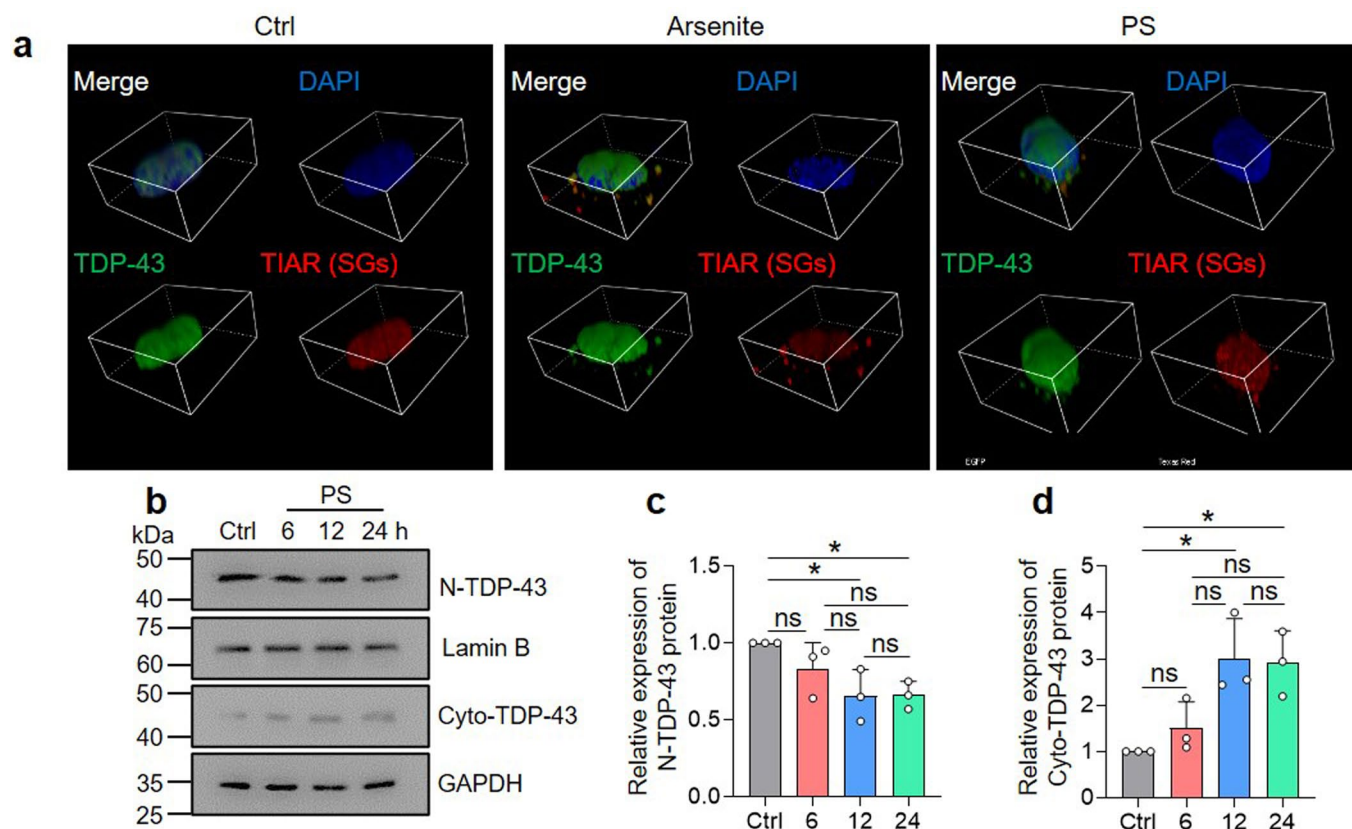
Extended Data Fig. 2 | PS exposure-induced cellular stress and compromised organelles in HeLa cells. **a**, Representative images of NAC inhibit PS-triggered nuclear ^{EGFP}TDP-43 condensates formation. NAC, 2 mM, PS, 100 μ g/mL, 6 h. **b**, The relative ROS levels (using the DCF-DA probe) were detected by flow cytometry. Data are expressed as means $\pm$ SD, $n=3$, as determined by one-way ANOVA with Tukey's multiple comparison tests (Ctrl vs. NAC, $p=0.9742$; Ctrl vs. PS, $p<0.0001$; Ctrl vs. PS + NAC, $p=0.1539$; PS vs. PS + NAC, $p<0.0001$). **c**, Quantification of nuclear ^{EGFP}TDP-43 condensates number *per cell* in (b) ($n=10$ cells). Data are displayed as the median $\pm$ the interquartile range, $n=3$, as determined by one-way ANOVA with Tukey's multiple comparison tests ($p<0.0001$; $ns, p>0.05$; Ctrl vs. PS+NAC, $p=0.0081$). **d**, Representative images of p-eIF2 α expression levels in HeLa cells. GAPDH was used as internal control. **e**, Quantification of p-eIF2 α expression in (d). Data are expressed as means $\pm$ SD, $n=3$, as determined by one-way ANOVA with Tukey's multiple comparison tests. (Ctrl vs. PS 6 h, $p=0.0404$;

Ctrl vs. PS 12 h, $p=0.0092$; Ctrl vs. PS 24 h, $p=0.0300$; $ns, p>0.05$). **f**, Representative images of JC-1, AO and Magic red staining in HeLa cells after PS exposure. Red JC-1 aggregates indicate high MMP, and the green JC-1 monomers indicate low MMP; Lysosomal membrane permeability was detected by AO. AO in the intact lysosomes (red), cytoplasm (green); Cathepsin B release was detected by Magic Red staining. Magic Red (red), DAPI (blue). $n=3$. **g**, Flow cytometry was used to quantify PS-induced MMP loss (JC-1), cell membrane damage (PI), lysosomal membrane permeability (AO) and Cathepsin B release (Magic Red). Data are expressed as means $\pm$ SD, $n=3$, as determined by one-way ANOVA with Tukey's multiple comparison tests (For JC-1 test, Ctrl vs. 6, 12, 24 h, $p=0.0075$, $p=0.0002$, $p<0.0001$, respectively; For PI test, Ctrl vs. 6, 12, 24 h, $p=0.0371$, $p<0.0001$, $p<0.0001$, respectively; For AO test, Ctrl vs. 6, 12, 24 h, $p=0.2378$, $p<0.0001$, $p<0.0001$, respectively; For Cathepsin B test, Ctrl vs. 6, 12, 24 h, $p=0.0003$, $p<0.0001$, $p<0.0001$, respectively).



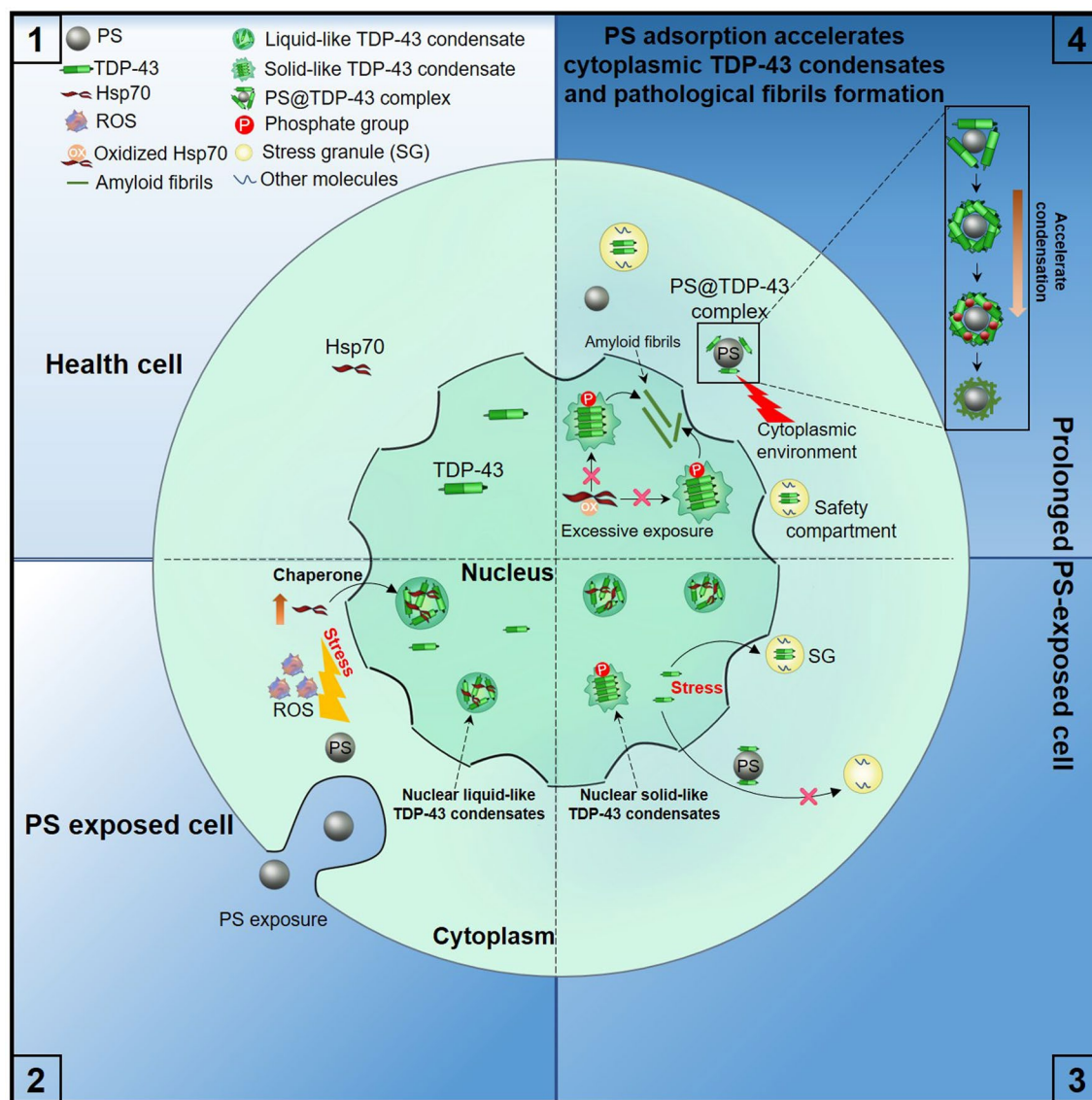
Extended Data Fig. 3 | Solid-like nuclear TDP-43 condensates formation is required for PS exposure-induced TDP-43 pathology. **a**, EGFP-TDP-43 transfected HeLa cells exposed to PS (100 μ g/mL) followed by immunostaining with anti-pTDP-43 (Ser409/410). Nuclei are marked by dashed circles. **b**, The percentage of cells expressing pTDP-43 in (a). Data are expressed as means $\pm$ SD ($n = 3$, 25 cells *per* experiment), as determined by one-way ANOVA with Tukey's multiple comparison tests (Ctrl *vs.* PS 12 h, $p = 0.0121$; Ctrl *vs.* PS 24 h, $p = 0.0010$; PS 6 h *vs.* PS 12 h, $p = 0.0314$; PS 6 h *vs.* PS 24 h, $p = 0.0021$; *ns*, $p > 0.05$). **c**, Representative images of phosphorylation levels of nuclear TDP-43 (N-pTDP-43). Lamin B was used as an internal control. **d**, Quantification of nuclear TDP-43 phosphorylation. Data are expressed as means $\pm$ SD ($n = 3$), as determined by one-sided one-way ANOVA with Tukey's multiple comparison tests (Ctrl *vs.* PS 6 h, $p = 0.8011$; Ctrl *vs.*

PS 12 h, $p = 0.0018$; Ctrl *vs.* PS 24 h, $p < 0.0001$; PS 6 h *vs.* PS 12 h, $p = 0.0051$; PS 6 h *vs.* PS 24 h, $p < 0.0001$; PS 12 h *vs.* PS 24 h, $p < 0.0001$). **e**, AmyT staining. Nuclei are marked by dashed circles. **f**, The percentage of cells showing AmyT in (e). Data are expressed as means $\pm$ SD ($n = 3$, 25 cells *per* experiment), as determined by one-sided one-way ANOVA with Tukey's multiple comparison tests (Ctrl *vs.* PS 24 h, $p = 0.0024$; PS 6 h *vs.* PS 24 h, $p = 0.0040$; PS 12 h *vs.* PS 24 h, $p = 0.0117$; *ns*, $p > 0.05$). **g**, Representative images of HeLa cells expressing WT or A326P TDP-43-HA in the presence of PS (100 μ g/mL). **h**, CCK8 assay showed the effect of PS-triggered TDP-43 LLPS on the viability of HeLa cells. Data are expressed as means $\pm$ SD ($n = 3$), as determined by two-way ANOVA with Bonferroni's multiple comparison tests (PS 6 h, WT *vs.* A326P, $p = 0.0153$; PS 12 h, WT *vs.* A326P, $p = 0.0069$; PS 24 h, WT *vs.* A326P, $p = 0.0465$; *ns*, $p > 0.05$).



Extended Data Fig. 4 | PS exposure triggers TDP-43 cytoplasmic translocation. a, Three-dimensional reconstruction images of TDP-43 recruitment to SGs in Fig. 3a. **b**, Representative images of TDP-43 expression levels in the nucleus (N-TDP-43) and the cytoplasm (Cyto-TDP-43) in HeLa cells. Lamin B and GAPDH were used as internal controls for the nucleus and cytoplasm, respectively. **c**, Quantification of TDP-43 expression in the nucleus.

Data are expressed as means $\pm$ SD, $n = 3$, as determined by one-way ANOVA with Tukey's multiple comparison tests (Ctrl vs. PS 12 h, $p = 0.0444$; Ctrl vs. PS 24 h, $p = 0.0464$; ns, $p > 0.05$). **d**, Quantification of TDP-43 expression in the cytoplasm. Data are expressed as means $\pm$ SD, $n = 3$, as determined by one-way ANOVA with Tukey's multiple comparison tests (Ctrl vs. PS 12 h, $p = 0.0189$; Ctrl vs. PS 24 h, $p = 0.0240$; ns, $p > 0.05$).



Extended Data Fig. 5 | PS exacerbates the pathological tendency of TDP-43. Schematic illustration of the role of PS exposure in the aberrant phase separation and pathology of TDP-43.

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Software and code

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Data collection	For imaging on Nikon confocal fluorescence microscope, NIS-Elements AR 5.01.00 64-bit software is used. For OLYMPUS IX71 fluorescence microscope, cellSens Standard software is used. For flow cytometry data collection, BD FACShorus software is used.
Data analysis	GraphPad 8.0.2, ImageJ, FlowJo v10, NIS-Elements AR Analysis 5.01.00 64-bit, 1.53c, Zeta View 8.05.04, Broad Institute GSEA, Ligplot+, Pymol and EthoVision XT were used to perform data analysis. No custom code/algorithm was used.

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Figures 1-5, 7 and Supplementary Figures 1, 2, 4-8, 12 and 15 have associated raw data upon request.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	All our in vitro experiments described in this paper were performed with $n > 3$. A minimum of 3 independent repeats was required to evaluate statistical significance. The in vivo studies were performed with $n > 3$ mice per group and minimizing the number of animals to still achieve statistical validity. The animal experiments were approved by the Institutional Animal Care and Use Committee of Southwest University (IACUC-20230313-04) following the National Institutes of Health guidelines. Details regarding sample sizes and statistical tests of all experiments are provided in the Methods section and figure captions.
Data exclusions	We did not exclude any data.
Replication	The experiments were repeated for 3 times in vitro, and the results were consistent with each other. For the in vivo study, each group contained 3 mice.
Randomization	Randomization was relevant in the mice experiments. And the mice were age-matched before randomization into the various treatment groups.
Blinding	The sub team working on the mouse experiments was independent from the sub team analyzing the mouse data, also was blinded from the experimental conditions/groups. No data were excluded.

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We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	TDP-43 Polyclonal antibody (Proteintech; Product number: 10782-2-AP), 1:1000 (WB), 1:500 (IF), dilution G3BP1 Polyclonal antibody (Proteintech; Product number: 13057-2-AP), 1:500 (IF), dilution TIA1 Polyclonal antibody (Proteintech; Product number: 12133-2-AP), 1:500 (IF), dilution Phospho-TDP43 (Ser409/410) Polyclonal antibody (Proteintech; Product number: 22309-1-AP), 1:1000 (WB), 1:500 (IF), dilution HSP70 Polyclonal antibody (Proteintech; Product number: 10995-1-AP), 1:5000 (WB), dilution HA tag Polyclonal antibody (Proteintech; Product number: 151064-2-AP), 1:100 (IF), dilution CoraLite488-conjugated Goat Anti-Rabbit IgG(H+L) (Proteintech; Product number: SA00013-2), 1:100 (IF), dilution Cy3-conjugated Affinipure Goat Anti-Rabbit IgG(H+L) (Proteintech; Product number: SA00009-2), 1:100 (IF), dilution Lamin B1 Polyclonal antibody (Proteintech; Product number: 12987-1-AP), 1:5000 (WB), dilution GAPDH Polyclonal antibody (Proteintech; Product number: 10494-1-AP), 1:5000 (WB), dilution
Validation	Validation of each antibody was performed under information offered by each supplier.

Eukaryotic cell lines

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Cell line source(s)	HeLa (CL-0101) was obtained from Procell Life Science & Technology Co., Ltd (Wuhan, China).
Authentication	We did not authenticate the cell lines because they were all from commercial sources and they had their own individual companies' quality control policies.
Mycoplasma contamination	We did not detect the mycoplasma contamination because the source company already tested and the results showed no mycoplasma in all cell lines.
Commonly misidentified lines (See ICLAC register)	None

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals	7-8 weeks old male C57BL/6J mice were obtained from the Hunan SJA Laboratory Animal Co., Ltd. (Hunan, China),
Wild animals	No wild animals were used in our study.
Field-collected samples	No field collected samples were used in the study.
Ethics oversight	The Institutional Animal Care and Use Committee of Southwest University (IACUC-20230313-04).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Flow Cytometry

Plots

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- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	HeLa cells (1×10^6 cells per well) were grown on 6-well plates and exposed to N-acetyl-L-cysteine (NAC) (2 mM) and/or PS (100 $\mu\text{g}/\text{mL}$) for 6 h. Cells in the control group received serum-free medium treatment for 6 h and were then incubated with DCFH-DA (Sigma-Aldrich, USA) probe in the dark at 37 °C for 30 min. Cells were washed three times with room temperature PBS and collected by centrifugation. HeLa cells (1×10^6 cells per well) were grown on 6-well plates. Fresh DMEM media with PS (100 $\mu\text{g}/\text{mL}$) was added to the plates for indicated time. Then cells were incubated with JC-1 probe (Yeasen Biotechnology, China) in dark at 37 °C for 15 min. Cells were washed three times with room temperature PBS and collected by centrifugation. HeLa cells (1×10^6 cells per well) were grown on 6-well plates. After treatment, cells were stained with acridine orange (AO) (5 $\mu\text{g}/\text{mL}$) (Amresco, USA) at 37 °C for 30 min for assessment of lysosomal permeability. Cells were washed three times with room temperature PBS and collected by centrifugation. HeLa cells (1×10^6 cells per well) were grown on 6-well plates. PS-challenged cells were stained with Magic Red (ImmunoChemistry Technologies, USA) solution (200 μL) at 37 °C for 30 min. Then cells were washed three times with room temperature PBS and collected by centrifugation. HeLa cells (1×10^6 cells per well) were grown on 6-well plates. After treatment, cells were stained with propidium iodide (PI) (2 μM) (Yeasen Biotechnology, China) at 37 °C for 30 min for analysis of cell membrane integrity. Then cells were washed three times with room temperature PBS and collected by centrifugation. HeLa cells (1×10^6 cells per well) were seeded in 6-well plates and incubated with a serum-free medium containing RFPS. Then, the cells were washed three times with PBS and collected by centrifugation. A total of 1×10^4 events was acquired for each sample from three independent experiments.
Instrument	BD FACS Melody flow cytometer.
Software	BD FACSCorus software was used for data collection and FlowJo v10 was used for data analysis
Cell population abundance	$>1 \times 10^4$ events were recorded for the desired population.
Gating strategy	FACS gating for measuring the cellular uptake of RFPS via PE channel. FACS gating strategy for measuring the ROS value of HeLa cells via FITC channel (cells were incubated by DCFH-DA indicator). FACS gating strategy for measuring the

mitochondrial membrane potential of HeLa cells via FITC and PE channel (cells were incubated by JC-1 probe). FACS gating strategy for measuring the lysosomal damage of HeLa cells via FITC channel (cells were incubated by acridine orange indicator). FACS gating strategy for measuring the cathepsin B release of HeLa cells via PE channel (cells were incubated by Magic Red indicator). FACS gating strategy for measuring the cathepsin B release of HeLa cells via PE channel (cells were incubated by Magic Red indicator). FACS gating strategy for measuring the cell membrane damage of HeLa cells via PE channel (cells were incubated by propidium iodide indicator).

☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.